

**KINETICS AND OPTIMISATION OF ENZYMATIC HYDROLYSIS OF STARCH
FROM CORN COB VIA RESPONSE SURFACE METHODOLOGY**

**AZUBUIKE, PROMISE LEONARD
18/53473U/2**

OCTOBER, 2025

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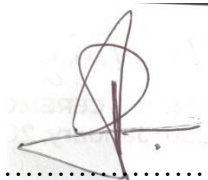
**A RESEARCH PROJECT REPORT SUBMITTED TO THE DEPARTMENT OF
CHEMICAL ENGINEERING, FOR THE FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF DEGREE OF BACHELOR OF
ENGINEERING IN CHEMICAL ENGINEERING**

OCTOBER, 2025

DECLARATION

I hereby declare that this work was done by me, under the supervision of Prof. S. Abdulsalam and it has never been presented anywhere for the award of a degree or a certificate. All sources have been duly acknowledged.

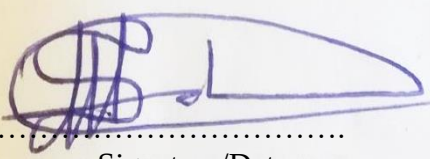
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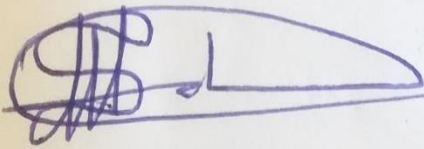
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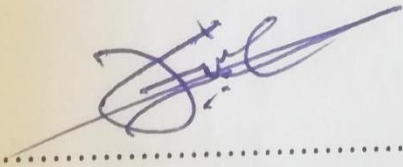
This project report entitled ***Kinetics and Optimisation of Enzymatic Hydrolysis of Starch from Corn Cob via Response Surface Methodology*** ' meets the regulations governing the award of the degree of Bachelor of Engineering of Abubakar Tafawa Balewa University, Bauchi and is approved for its contribution to knowledge and literary presentation.



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DEDICATION

I dedicate this project work to my most beloved parents in the persons of Mr. and Mrs. Paul Idu for their immense contribution and support throughout my life and to my beloved siblings.

ACKNOWLEDGEMENTS

My profound gratitude goes to Almighty God, for giving me the strength, courage, endurance and finally seeing me through this work. This acknowledgment will be incomplete if I forget to acknowledge Prof. S. Abdulsalam, my mentor, lecturer as well as my project supervisor for scheduling his time, out of no time in order to give me advice, make corrections and comments towards this project.

I equally want to express my deep appreciation to my parents Chief and Mrs. Paul Idu for their financial support and prayer throughout my study. My siblings Uche, Ada, Nonye and Caroline are also supportive to the success of this study. All members of the Material and Energy Laboratory you all have been very supportive. I cannot forget my Lecturers and course mates/friends, my reading group BNALN for their support.

ABSTRACT

The increasing demand for renewable energy and bio-based products has heightened the need for sustainable utilization of agricultural residues, among which corn cob is abundant and rich in starch that can serve as a low-cost feedstock for fermentable sugar production. Despite its potential, the enzymatic hydrolysis of corn cob starch is often limited by suboptimal process conditions and inadequate kinetic data, which restrict large-scale application. This study therefore investigated the kinetics and optimisation of enzymatic hydrolysis of starch extracted from corn cob using α -amylase. Response Surface Methodology (RSM) was employed to evaluate the combined effects of enzyme concentration, pH, and reaction time on glucose yield. Among the models tested, the reduced cubic model was selected because it provided the best statistical performance, with high R^2 (0.9966) and adjusted R^2 (0.9907), significant model terms, and a non-significant lack of fit, making it the most reliable predictor of glucose yield. Experimental results showed that enzyme concentration and hydrolysis time were the most significant factors, with optimum conditions achieved at 5.596 mg/mL enzyme concentration, pH 6.117, and 89.667 minutes, yielding 99.61% glucose. Validation runs confirmed the accuracy of the model with experimental yields ranging from 98.6% to 98.9%. Kinetic analysis revealed that the hydrolysis followed a fractional reaction order ($n = 0.56$), while Michaelis–Menten parameters were determined as $V_{max} = 0.368$ mg/mL·min and $K_m = 10.44$ mg/mL, indicating efficient catalysis and good enzyme–substrate affinity. FTIR analysis of the hydrolysate confirmed glucose formation through characteristic O–H, C–O, and anomeric absorption bands. These findings demonstrate that α -amylase can effectively hydrolyze corn cob starch and that RSM, with the reduced cubic model, is a robust and reliable tool for process optimisation.

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ABBREVIATIONS, DEFINITIONS AND SYMBOLS

RSM	Response Surface Methodology
CCD	Central Composite Design
BBD	Box-Behnken Design
FTIR	Fourier Transform Infrared Spectroscopy
DNS	Dinitrosalicylic Acid
ANOVA	Analysis of Variance
DH	Degree of Hydrolysis
$V_{\max}$	Maximum Reaction Velocity
K_m	Michaelis Constant
SEM	Scanning Electron Microscopy
DSC	Differential Scanning Calorimetry

Symbol	Definition	Unit
A	Enzyme concentration	mg/mL
B	pH	–
C	Reaction time	Min
k_1	First-order rate constant	min^{-1}
k_2	Second-order rate constant	$\text{L} \cdot \text{mg}^{-1} \cdot \text{min}^{-1}$
K_m	Michaelis constant	mg/mL
$V_{\max}$	Maximum reaction velocity	$\text{mg/mL} \cdot \text{min}$
Y	Glucose yield	%
T	Reaction time	min
$[C_0]$	Initial substrate concentration	mg/mL
$[C]$	Substrate concentration at time t	mg/mL

CHAPTER ONE

1.0

INTRODUCTION

1.1 Background of the Study

The growing worldwide interest in bio-based products and ecologically oriented manufacturing procedures has been provoking a research interest in technologies for processing biomasses. Of several biomass types worth attention, corn cobs (an agricultural residue) have received much interest since they are rich in starch that can be hydrolyzed to fermentable sugars to make bioethanol, food processing and pharmaceutical applications (Zhang *et al.*, 2020). Enzymatic hydrolysis of starch is a major step in this conversion process through an environmental friendly and efficient way as compared to the traditional acid hydrolysis methods (Li *et al.*, 2022).

Enzymatic hydrolysis refers to the process by which starch is digested into simpler sugars by amylolytic enzymes such as α amylase and glucoamylase. This process is affected by various factors such as temperature, pH, enzyme concentration, substrate concentration and reaction period, all these factors need to be optimized to effectively enhance the efficiency of hydrolysis (Chen *et al.*, 2023). Kinetic aspects of this reaction have to be well understood for the enhancement of process efficiency and scaling up on the application in industry. First order, second order kinetic models and Michaelis-Menten equations are often used to explain starch hydrolysis enzymatic's rate of reaction, but they revealed much about the mechanisms of reaction (Wang *et al.*, 2020).

Response surface methodology (RSM) is a statistical method for solving multivariate factors. Specific data are obtained through the test, and a polynomial quadratic equation is used to fit a functional relationship between the variables and response. The fitted

polynomial equation is analyzed to find the optimum process parameters (Wang *et al.* 2020). The response surface methodology (RSM) has come out strongly as a potent statistical tool for optimizing biochemical processes. It allows in simultaneous multiple process parameters assessment with the aim of determining favorable conditions for maximal starch hydrolysis processing yield (Yang & Liu, 2021). Through RSM, researchers are able to generate predictive models to set, fine tuning for reaction condition, minimize experimental cost and increase overall process efficiency (Sun *et al.*, 2024).

1.2 Problem Statement

Corn cobs, almost ubiquitous agricultural residue, present great promise as a renewable agricultural waste feedstock for the enzymatic hydrolysis production of fermentable sugars. Optimisation of the process of hydrolysis of corn cob to obtain maximum efficiency, however, remains one of the major challenges to be addressed because of the complex relationship of process variables such as' enzyme concentration, temperature, pH and reaction time. Studies on starch hydrolysis from different biomass sources have been conducted previously, but there is a deficit of holistic research on a kinetic model and optimisation of the process of enzymatic starch hydrolysis from corn cobs in particular.

1.3 Aim & Objectives

1.3.1 Aim

The aim of this study is to investigate the kinetics and optimisation of enzymatic hydrolysis of starch from corn cob via response surface methodology.

1.3.2 Objectives

The objectives of this study are to:

1. Recover starch from corn cob and hydrolyze it enzymatically using α -amylase.
2. Apply Response Surface Methodology (RSM) to optimize key process variables (Enzyme concentration, pH and time).
3. Model the enzymatic hydrolysis kinetics using 1st order, 2nd order and shifting order kinetic equations.

1.4 Significance of Study

This study offers important insights into efficiency of starch hydrolysis from corn cob. Understanding the kinetic parameters ruling the enzymatic hydrolysis process is also significant because it participates in the development of sustainable biomass utilization, promoting the optimisation of the enzymatic hydrolysis of starch from the corn cobs. Using a thorough analysis of hydrolysis kinetics, and by employing Response Surface Methodology (RSM) for process optimisation, this research generates useful insights to facilitate better industrial use of enzymatic hydrolysis of starch.

1.5 Scope and Limitations

This study is concerned with the enzymatic hydrolysis of starch from corn cobs with a special emphasis on enzymatic process kinetics and its process optimisation using Response Surface Methodology (RSM). The experimental work was also carried out with special amylolytic enzyme such as α -amylase to hydrolyze starch and determine starch-conversion efficiency under different conditions. It also, covers the kinetic modeling of the enzymatic reaction of Corn cobs. The findings will only be confined to hydrolysis of starch from corn cob through the use of only enzymatic hydrolysis.

CHAPTER TWO

2.0 LITERATURE REVIEW

2.1 Introduction

The enzymatic hydrolysis of starch from corn cobs has garnered significant attention due to its potential in producing fermentable sugars for bioethanol production and other industrial applications. This study focuses on the enzymatic hydrolysis of starch from corn cob using α -amylase, a key enzyme for starch liquefaction, and the application of Response Surface Methodology (RSM) for process optimisation. The review covers starch extraction from corn cob, the role of α -amylase in hydrolysis, kinetic modeling, and RSM optimisation studies, while identifying research gaps specific to the use of α -amylase for corn cob starch hydrolysis.

2.2 Corn Cob Characteristics and Chemical Composition

Corn cob is the hard, central part of the maize ear that remains after the grains have been removed. It is one of the common agricultural wastes in many parts of the world, including Nigeria. Chemically, corn cob is a lignocellulosic material, meaning it is made up of cellulose, hemicellulose, lignin, starch, and a small amount of ash. The exact composition depends on the maize variety and growing conditions.

On average, corn cob contains about 33–45% cellulose, 26–37% hemicellulose, 17–22% lignin, 5–10% starch, and 1–2% ash (Zhang *et al.*, 2020; Sewsynker-Sukai *et al.*, 2023; Li *et al.*, 2024). Cellulose and hemicellulose are the main carbohydrate sources and are useful for producing sugars after hydrolysis. Lignin, on the other hand, gives the cob its strength but also makes it harder for enzymes to access the carbohydrates. Starch is present in

smaller amounts compared to maize grains, but it can still be used for sugar production.

Table 2.1 shows the typical composition of corn cob as reported in recent studies.

Table 2.1: Composition of Corn Cob (dry basis)

Component	Range (%)	References
Cellulose	33 – 45	Zhang <i>et al.</i> , 2020; Sewsynker-Sukai <i>et al.</i> , 2023
Hemicellulose	26 – 37	Kumar <i>et al.</i> , 2022; Li <i>et al.</i> , 2024
Lignin	17 – 22	Sewsynker-Sukai <i>et al.</i> , 2023; Chen <i>et al.</i> , 2022
Starch	5 – 10	Zhang <i>et al.</i> , 2020; Li <i>et al.</i> , 2024
Ash	1 – 2	Maghfiroh <i>et al.</i> , 2021

The functional groups in corn cob are usually identified using Fourier Transform Infrared Spectroscopy (FTIR). This method shows the different bonds and structures present in the material. A broad peak at around 3400 cm^{-1} indicates --OH groups from cellulose, hemicellulose, and lignin. The band near 2920 cm^{-1} comes from C--H stretching in carbohydrates. The peak around 1730 cm^{-1} is linked to carbonyl groups in hemicellulose, while peaks between 1600 and 1515 cm^{-1} show the aromatic structures of lignin. A peak near 1430 cm^{-1} is related to crystalline cellulose, and strong bands in the $1050\text{--}1100\text{ cm}^{-1}$ region show glycosidic linkages in cellulose and hemicellulose. Finally, a peak at about 897 cm^{-1} is typical for β -glycosidic bonds in cellulose (Maghfiroh *et al.*, 2021; Chen *et al.*, 2022; Kumar *et al.*, 2022). These FTIR features are summarized in Table 2.2.

Table 2.2: FTIR Peaks of Corn Cob

Wavenumber (cm ⁻¹)	Functional Group	Component	References
~3400	O–H stretching	Cellulose, hemicellulose, lignin	Maghfiroh <i>et al.</i> , 2021; Kumar <i>et al.</i> , 2022
~2920	C–H stretching	Cellulose, hemicellulose	Chen <i>et al.</i> , 2022
~1730	C=O stretching	Hemicellulose	Sewsynker-Sukai <i>et al.</i> , 2023
1600 – 1515	Aromatic C=C	Lignin	Zhang <i>et al.</i> , 2020
~1430	CH ₂ scissoring (crystallinity)	Cellulose	Li <i>et al.</i> , 2024
1380 – 1370	C–H bending	Cellulose, hemicellulose	Chen <i>et al.</i> , 2022
~1155	C–O–C stretching	Polysaccharides	Kumar <i>et al.</i> , 2022
1050 – 1030	C–O stretching	Cellulose, hemicellulose	Maghfiroh <i>et al.</i> , 2021
~897	β-glycosidic bonds	Cellulose	Li <i>et al.</i> , 2024

2.3 Starch Extraction from Corn Cob

Starch extraction from corn cob typically involves wet milling or enzymatic pretreatment to separate starch from non-starch components such as proteins and fibers. Yang *et al.* (2019) reported a starch purity of 98.5% from corn cob using alkaline pretreatment followed by protease hydrolysis, which enhances starch accessibility for subsequent enzymatic processes. The extracted starch has an amylose content of 20–28%, with granular morphology and thermal properties similar to maize starch, as confirmed by Scanning Electron Microscopy (SEM) and Differential Scanning Calorimetry (DSC) (Wang *et al.*, 2020). However, studies on corn cob starch extraction often focus on

pretreatment for lignocellulose hydrolysis rather than optimizing conditions specifically for starch yield, limiting data on starch-specific extraction efficiency.

2.4 Enzymatic Hydrolysis of Starch

Starch, a polysaccharide composed of amylose and amylopectin, is a primary energy reserve in plants. Its hydrolysis involves breaking down the glycosidic bonds to yield simpler sugars like glucose. Enzymatic hydrolysis is a biochemical process in which enzymes catalyze the breakdown of complex macromolecules into simpler compounds in the presence of water (Kumar *et al.*, 2022). This process plays a crucial role in various industries, including food processing, biofuel production, and pharmaceuticals, where it enhances the efficiency of biochemical transformations and improves product yield (Singh *et al.*, 2020). Enzymatic hydrolysis offers several advantages over chemical hydrolysis, including greater specificity, lower energy requirements, and reduced formation of unwanted byproducts (Gurung *et al.*, 2022).

2.4.1 Types of enzymes

There are several types of enzymes used for enzymatic hydrolysis, including:

2.4.1.1 α Amylases

α -Amylase (EC 3.2.1.1) is an endo-acting enzyme that randomly cleaves α -1,4-glycosidic bonds in starch, producing dextrins and oligosaccharides during the liquefaction phase of hydrolysis (Singh *et al.*, 2017). Unlike glucoamylase, which is typically used for saccharification to produce glucose, α -amylase alone results in a mixture of reducing sugars (e.g., maltose, maltotriose) suitable for specific applications like syrup production or as intermediates for further processing. Kumar *et al.* (2018) studied α -amylase hydrolysis of maize starch, achieving a degree of hydrolysis (DH) of 85% at 60°C, pH 6.5,

and 2.5 U/mg enzyme concentration. For corn cob, the lignocellulosic matrix poses a challenge, requiring pretreatment to expose starch for enzymatic action. Chen *et al.* (2021) reported that pretreatment with aqueous ammonia improved enzyme accessibility in corn cob, but their study focused on cellulase-based hydrolysis rather than α -amylase.



Plate 2.1 α Amylase Enzyme (EC 3.2.1.1)

2.4.1.2 Glucoamylases

These enzymes specifically break down starch into glucose monomers. They are commonly used in fermentation industries to convert starch-rich substrates into fermentable sugars for ethanol production (Kumar *et al.*, 2022).

2.4.1.3 Proteases

Proteases catalyze the hydrolysis of proteins into peptides and amino acids. They have extensive applications in food processing, pharmaceuticals, and detergent manufacturing (Singh *et al.*, 2020).

2.4.1.4 Cellulases

These enzymes break down cellulose into glucose units. They are essential in biofuel production, as they convert plant biomass into fermentable sugars (Kumar *et al.*, 2022).

2.4.1.5 Lipases

Lipases catalyze the hydrolysis of lipids into glycerol and free fatty acids. They are widely used in biodiesel production, dairy processing, and pharmaceutical formulations (Gurung *et al.*, 2022).

2.4.2 Factors affecting enzymatic hydrolysis

Enzymatic hydrolysis is a complex biochemical process influenced by various parameters that affect the efficiency and extent of substrate conversion. Understanding these factors is crucial for optimizing hydrolysis conditions, particularly when dealing with lignocellulosic biomass or starch-based feedstocks such as corn cobs. The key factors influencing enzymatic hydrolysis include:

2.4.2.1 Temperature

Temperature plays a crucial role in determining the rate and efficiency of enzymatic hydrolysis. Enzymes have a specific temperature range within which they exhibit optimal catalytic activity. Typically, enzymatic activity increases with temperature due to enhanced kinetic energy, which improves molecular collisions between enzyme and substrate. However, beyond the optimal range (often between 40°C and 60°C for most commercial enzymes), enzyme structure begins to destabilize, leading to denaturation, loss of active site integrity, and subsequent decline in catalytic performance. For instance, cellulases and amylases commonly used in biomass hydrolysis are thermolabile and show peak activity

around 50°C (Gurung *et al.*, 2022). Sustaining optimal temperature is thus vital for maximizing hydrolysis efficiency while preventing enzyme degradation.

2.4.2.2 pH

The pH of the reaction medium significantly influences enzyme structure and function. Each enzyme possesses an optimal pH at which its active site configuration is most compatible with substrate binding and catalysis. Deviations from this optimal pH can lead to ionization of amino acid residues, particularly those in the active site, which affects enzyme-substrate affinity and the overall reaction rate. For example, amylase performs optimally around pH 5.5–6.5, while cellulase exhibits peak activity near pH 4.5–5.5. Extreme acidic or alkaline environments can cause irreversible denaturation of enzymes and decrease hydrolysis efficiency (Kumar *et al.*, 2022).

2.4.2.3 Substrate Concentration

Substrate concentration directly influences the rate of enzymatic hydrolysis through the Michaelis–Menten kinetics framework. Initially, increasing substrate concentration leads to a proportional increase in reaction rate. However, once the enzyme's active sites become saturated (maximum velocity), $V_{\max}$, further increases in substrate concentration do not enhance the rate, as the enzymes are operating at full capacity. In addition, excessive substrate levels may cause viscosity issues or mass transfer limitations, which hinder enzyme-substrate interactions (Singh *et al.*, 2020). Therefore, determining the optimal substrate loading is essential for efficient bioconversion without causing process inhibition.

2.4.2.4 Enzyme Concentration

The enzyme concentration is another critical parameter in hydrolysis. A higher enzyme dosage typically results in faster hydrolysis due to an increased number of available active sites for catalysis. However, this relationship holds true only when the substrate is in excess. Beyond a certain point, increasing enzyme concentration does not significantly affect hydrolysis rate due to substrate limitation. Moreover, higher enzyme loads may not be economically viable, especially in industrial-scale operations. Thus, optimizing enzyme loading is key to balancing cost and performance (Kumar *et al.*, 2022).

2.4.2.5 Presence of Inhibitors

Inhibitory compounds, whether present naturally in the biomass or formed as hydrolysis byproducts, can negatively impact enzymatic performance. Common inhibitors include phenolic compounds, organic acids (like lactic and acetic acid), heavy metals, and end-products such as glucose or cellobiose, which may exhibit feedback inhibition. These inhibitors may alter enzyme conformation, obstruct the active site, or interfere with substrate binding. For example, glucose accumulation during hydrolysis can inhibit β -glucosidase, reducing overall saccharification efficiency (Gurung *et al.*, 2022). Strategies such as detoxification, pH adjustment, or using enzyme cocktails can help mitigate these effects.

2.4.2.6 Reaction Time

The duration of enzymatic hydrolysis impacts the extent of substrate degradation and the yield of fermentable sugars. Generally, longer reaction times lead to greater hydrolysis. However, overextended hydrolysis can lead to enzyme inactivation, accumulation of inhibitory compounds, and increased processing costs. It has been observed that most

enzymatic conversions reach their peak yield within the first 24–48 hours, after which the rate levels off due to product inhibition and reduced enzyme activity (Singh *et al.*, 2020). Hence, optimizing the hydrolysis duration is important to maximize yield while maintaining economic feasibility.

2.4.3 Recent advances in enzymatic hydrolysis applications

Recent studies have focused on improving enzymatic hydrolysis efficiency and expanding its industrial applications. Some notable areas of research include:

2.4.3.1 Biofuel Production

Enzymatic hydrolysis plays a crucial role in biofuel production by converting lignocellulosic biomass into fermentable sugars. Recent advancements in enzyme engineering and pretreatment methods have enhanced the efficiency of cellulase and hemicellulase enzymes, leading to higher bioethanol yields (Kumar *et al.*, 2022). Researchers have also explored genetically modified microorganisms that produce highly efficient hydrolytic enzymes for more cost-effective biofuel production (Singh *et al.*, 2020).

2.4.3.2 Food Processing

Enzymatic hydrolysis is widely used in the food industry to enhance the nutritional and functional properties of food products. Amylases and glucoamylases are used to hydrolyze starch for the production of syrups and maltodextrins, while proteases are utilized to improve protein digestibility in dairy and meat products (Gurung *et al.*, 2022). Recent advancements in enzyme immobilization techniques have improved enzyme stability and reusability, making enzymatic hydrolysis more economically viable in food processing (Kumar *et al.*, 2022).

2.4.3.3 Pharmaceutical Applications

Enzymatic hydrolysis is used in the pharmaceutical industry to enhance drug bioavailability and facilitate drug delivery. For instance, lipases are used to improve the solubility of poorly water-soluble drugs, while proteases aid in the hydrolysis of protein-based therapeutics for controlled drug release (Singh *et al.*, 2020). Advances in enzyme biotechnology have led to the development of novel biocatalysts with enhanced specificity and stability for pharmaceutical applications (Gurung *et al.*, 2022).

2.4.3.4 Biodegradable Plastics

Enzymatic hydrolysis has been explored as a sustainable method for breaking down biodegradable plastics into monomers that can be reused in polymer production. Researchers are investigating engineered enzymes that can efficiently degrade synthetic polymers like polyethylene terephthalate (PET) and polylactic acid (PLA) under mild conditions (Kumar *et al.*, 2022).

2.4.3.5 Waste Management and Environmental Remediation

Enzymatic hydrolysis is being utilized for the bioconversion of agricultural and industrial waste into value-added products. Enzymatic treatment of wastewater, for example, helps break down organic pollutants and improve water quality (Singh *et al.*, 2020). Enzyme-based hydrolysis methods have also been employed in the degradation of lignocellulosic residues to produce bio-based chemicals (Gurung *et al.*, 2022).

2.4.4 Mechanism of enzymatic hydrolysis

Enzymatic hydrolysis follows a catalytic mechanism in which enzymes bind to a specific substrate, facilitate its cleavage, and release the resulting smaller molecules. This process involves several steps:

2.4.4.1 Substrate Binding

Enzymatic hydrolysis begins with the recognition and binding of the substrate to a specialized region on the enzyme known as the active site. This process is governed by molecular complementarity, where the geometric and chemical properties of the active site align precisely with those of the substrate. Two main models explain this interaction: the lock-and-key model, which suggests that the substrate fits perfectly into the enzyme's active site without structural change, and the induced-fit model, which proposes that the enzyme undergoes a conformational adjustment upon substrate interaction to achieve an optimal fit (Voet & Voet, 2020). The binding process is further stabilized by non-covalent interactions such as hydrogen bonding, hydrophobic forces, electrostatic interactions, and van der Waals forces. This specificity ensures that enzymes selectively catalyze particular reactions, which is crucial for industrial bioprocesses where undesired side reactions must be minimized.

2.4.4.2 Catalysis

Once the enzyme-substrate complex is formed, the enzyme facilitates the hydrolysis reaction by providing an alternative reaction pathway with a lower activation energy. In hydrolytic enzymes such as amylases, cellulases, and proteases, water molecules are directly involved in the cleavage of glycosidic, peptide, or ester bonds. The catalytic residues in the active site (such as acidic or basic amino acids) act as proton donors or

acceptors, stabilizing transition states and promoting nucleophilic attack on the substrate bonds (Nelson & Cox, 2021). This precise orchestration of molecular events ensures that bond cleavage occurs efficiently and under mild conditions, which is a significant advantage compared to harsh chemical hydrolysis.

2.4.4.3 Product Release

Following catalysis, the hydrolyzed products are released from the active site into the surrounding medium. This step is often driven by the reduced affinity of the enzyme for the reaction products compared to the substrate. For instance, in starch hydrolysis, glucose or maltose molecules diffuse away from the active site once formed. The release of products is a critical step because it regenerates the free active site, enabling the enzyme to bind new substrate molecules rapidly. Product release is also influenced by environmental factors such as pH and ionic strength, which can affect enzyme-product interactions.

2.4.4.4 Enzyme Recycling

Enzymes emerge from the catalytic cycle structurally and functionally unaltered, enabling them to participate in multiple successive reactions. This catalytic turnover is a defining feature of enzymes, with some possessing exceptionally high turnover numbers (k_{cat}), meaning they can process thousands of substrate molecules per second. In industrial applications, such as starch hydrolysis for bioethanol production or glucose syrup manufacture, this property significantly enhances process efficiency and reduces the overall enzyme requirement (Nelson & Cox, 2021). Furthermore, enzyme immobilization technologies are increasingly applied to facilitate enzyme recovery and reuse, extending their operational lifespan in continuous processes.

2.5 Kinetic Models for Starch Hydrolysis

Kinetic models are critical for understanding and optimizing enzymatic hydrolysis of starch, describing reaction rates and product yields under varying conditions such as enzyme concentration, pH, substrate concentration, and temperature. These models aid in predicting reaction behavior, facilitating the design of efficient hydrolysis processes for applications in biofuel, food, and pharmaceutical industries. This study focuses on enzymatic hydrolysis of corn cob starch using α -amylase, and employs three kinetic models: pseudo-first-order, pseudo-second-order, and Michaelis- Menten to characterize the reaction dynamics and optimize glucose yield (Haviland *et al.*, 2021).

2.5.1 First-order kinetics

First-order kinetics assumes that the rate of hydrolysis depends linearly on the concentration of the reactant (starch). This model is applicable when the enzyme is in sufficient excess, so the substrate concentration primarily governs the reaction. The differential form of the first-order rate law is shown in Equation (2.1):

$$\frac{d[C]}{dt} = -k_1[C] \quad \dots(2.1)$$

where:

- $[C]$ is the starch concentration (g/L or mol/L),
- k_1 is the pseudo-first-order rate constant (min^{-1}),
- $\frac{d[C]}{dt}$ is the rate of starch hydrolysis.

Integration of Equation (2.1) gives Equation (2.2):

$$\ln \frac{C_o}{C} = -k_1 t \quad \dots (2.2)$$

where $[C_0]$ is the initial starch concentration, and t is time (min). A plot of $\ln \frac{C_o}{C}$ versus t gives a straight line with a slope of $-k_1$, allowing experimental determination of the rate constant. This model is useful for initial rate studies in enzymatic hydrolysis when substrate depletion is minimal (Murray *et al.*, 2018).

2.5.2 Second-order kinetics

Second-order kinetics assumes that the rate of hydrolysis is proportional to the square of the substrate concentration or involves bimolecular interactions between substrate molecules and enzyme. The general rate law is expressed in Equation (2.3):

$$\frac{d[C]}{dt} = -k_2 [C]^2 \quad \dots (2.3)$$

where:

- $[C]$ is the starch concentration (g/L or mol/L),
- k_2 is the pseudo-second-order rate constant (L/g · min or L/mol · min),
- $\frac{d[C]}{dt}$ is the rate of starch hydrolysis.

Integrating equation (2.3) with the initial condition $[C] = [C_o]$ at $t = 0$, we obtain equation (2.4):

$$\frac{1}{[C]} = \frac{1}{[C_o]} + k_2 t \quad \dots (2.4)$$

A plot of $\frac{1}{[C]}$ versus t yields a straight line with slope of k_2 and intercept of $\frac{1}{[C_o]}$. This model is particularly relevant for conditions in this study (Section 3.4) where substrate

concentration is fixed at 5%w/v, and enzyme concentration varies, potentially mimicking pseudo-second-order behavior (Gupta *et al.*, 2019).

2.5.3 Michaelis-Menten Kinetics

The Michaelis-Menten model is widely used to describe enzyme-catalyzed reactions, including starch hydrolysis by α -amylase, due to its ability to account for substrate saturation effects. This model assumes the formation of an enzyme-substrate complex as a key step and it is represented in Eqn (2.5):



where:

- E is the enzyme (α -amylase),
- S is the substrate (starch),
- ES is the enzyme-substrate complex,
- P is the product (glucose),
- k_1 , k_{-1} , and k_2 are rate constants.

This model forms the basis for deriving the Michaelis-Menten equation under the quasi-steady-state approximation.

The reaction rate (v) is described by the Michaelis-Menten equation in Eqn (2.6):

$$v = \frac{V_{\max}[S]}{K_m + [S]} \quad \dots (2.6)$$

where:

- v is the reaction rate (mol/L · min),
- $V_{\max}$ is the maximum reaction velocity (mol/L · min) when all enzyme sites are saturated,
- K_m is the Michaelis constant (mol/L), the substrate concentration at which $v = V_{\max}/2$,
- $[S]$ is the substrate concentration (mol/L).

At low $[S]$, the reaction follows first-order kinetics ($v \propto [S]$), while at high $[S]$, it approaches zero-order kinetics ($v \approx V_{\max}$). This model is applied to determine kinetic parameters for starch hydrolysis.(Michaelis Menten, 1913).

2.5.3.1 Experimental Verification

The Michaelis-Menten model is validated experimentally by measuring initial reaction rates at varying substrate concentrations. Several graphical and computational methods can be used to estimate the kinetic constants K_m and $V_{\max}$.

2.5.3.1.1 Lineweaver-Burk Plot (Double Reciprocal Plot)

This method involves transforming the Michaelis-Menten equation into a linear form in Eqn (2.7):

$$\frac{1}{v} = \frac{K_m}{V_{\max}} * \frac{1}{[S]} + \frac{1}{V_{\max}} \quad \dots (2.7)$$

Plot $\frac{1}{v}$ against $\frac{1}{s}$

Slope = $K_m/V_{\max}$

Intercept = $1/V_{\max}$

Limitation: Magnifies errors at low substrate concentrations

2.5.3.1.2 Eadie-Hofstee Plot

Another linear transformation of the Michaelis-Menten equation is the Eadie-Hofstee form, obtained by rearranging as shown in Eqn (2.8):

$$v = -K_m \left(\frac{v}{[S]} \right) + V_{max} \quad \dots (2.8)$$

A Plot of v against $\frac{v}{[S]}$ yields a straight line with Slope = $-K_m$, and intercept Intercept = V_{max}

This method avoids the reciprocal of v and reduces error propagation.

2.5.3.1.3 Hanes-Woolf Plot

The Hanes-Woolf method rearranges the Michaelis-Menten equation into Eqn (2.9):

$$\frac{[S]}{v} = \frac{[S]}{V_{max}} + \frac{K_m}{V_{max}} \quad \dots (2.9)$$

A plot of $[S]/v$ versus $[S]$ provides a linear graph where:

$$\text{Slope} = \frac{1}{V_{max}} \text{ and intercept} = \frac{K_m}{V_{max}}$$

This method is often preferred because it gives a more even distribution of error across data points.

2.6 Optimisation Techniques: Response Surface Methodology (RSM)

Optimizing enzymatic hydrolysis involves fine-tuning multiple parameters, such as enzyme concentration, temperature, pH, and reaction time. Response Surface Methodology (RSM) is a statistical tool that evaluates the relationships between these variables and the response (e.g., hydrolysis yield). Response Surface Methodology (RSM) is a statistical and

mathematical technique used for modeling and optimizing complex processes. It was initially developed by Box and Wilson in 1951 for optimizing chemical processes but has since been widely applied in various fields, including engineering, food processing, pharmaceuticals, and environmental science (Myers *et al.*, 2021).

RSM is particularly useful when dealing with multiple input variables that influence a response variable. It helps in understanding interactions between factors, identifying optimal conditions, and reducing experimental costs.

Among the various experimental designs in RSM, Central Composite Design (CCD) and Box-Behnken Design (BBD) are two of the most widely applied due to their efficiency in fitting quadratic models. These designs help researchers explore interactions between factors and determine optimal conditions for a process.

2.6.1 Central composite design (CCD)

Central Composite Design (CCD) is a flexible and robust experimental design widely used in Response Surface Methodology (RSM) because of its ability to effectively estimate quadratic response surfaces. It is composed of three main elements: factorial points, which form the base of the experiment and can be full or fractional; axial (or star) points, which are positioned at a distance from the center to allow estimation of curvature; and center points, which are repeated runs at the midpoint of all factors to improve experimental accuracy and reduce error. Depending on the placement of the axial points, CCD can be classified into three types. The Circumscribed CCD (CCC) has axial points that extend beyond the factorial cube ($\alpha > 1$), the Inscribed CCD (CCI) places the axial points within the factorial space ($\alpha < 1$), while the Face-Centered CCD (CCF) positions the axial points at the faces of the cube ($\alpha = 1$). This structure allows CCD to capture both linear and quadratic effects while ensuring reliability and flexibility in optimization studies.

2.6.1.1 Advantages of CCD

The Central Composite Design (CCD) offers several advantages in optimization studies. It provides accurate predictions for quadratic models, making it highly suitable for processes with curvature effects. CCD also allows flexibility in choosing design parameters to ensure rotatability, which improves the uniformity of prediction across the design space. In addition, it effectively detects both factor interactions and curvature effects, making it a robust tool for modeling and optimization in complex experimental systems.

2.6.1.2 Limitations of CCD

Despite its strengths, the Central Composite Design (CCD) also has some limitations. It generally requires a large number of experimental runs, which can increase the cost, time, and resources needed for the study. In addition, the use of axial points may sometimes result in impractical or unrealistic experimental conditions, particularly when dealing with sensitive chemical or biochemical processes where extreme factor levels are not feasible.

2.6.2 Box-Behnken design (BBD)

The Box-Behnken Design (BBD) is an alternative to the Central Composite Design (CCD), offering a more efficient design that requires fewer experimental runs. Unlike CCD, which includes all factorial points, BBD places its design points at the midpoints of the edges of the experimental space as well as at the center of the design space to improve model precision. A key advantage of BBD is that it avoids extreme factor levels, thereby reducing experimental risks that may arise from hazardous or unstable conditions.

2.6.2.1 Advantages of BBD

The Box-Behnken Design (BBD) offers several advantages in experimental optimization. It requires fewer experimental runs compared to the Central Composite Design (CCD), making it more economical in terms of time and resources. In addition, BBD avoids the use of extreme factor levels, which reduces the risk of experimental errors and ensures safer operating conditions. It is also efficient for quadratic modeling, providing reliable results while minimizing overall experimental costs.

2.6.2.2 Limitations of BBD

Despite its efficiency, the Box-Behnken Design (BBD) also has some limitations. It is generally less effective than the Central Composite Design (CCD) in fitting full quadratic models, particularly when strong curvature effects exist in the system. This may reduce the accuracy of predictions in processes where higher-order interactions are significant. Another drawback is that BBD does not explore the extreme boundaries of the factor space, meaning that information about system behavior under extreme conditions is not captured. While this feature improves safety and practicality, it may limit the design's ability to fully characterize responses in some biochemical or chemical processes where boundary conditions are important.

2.6.3 Comparison of CCD and BBD

In the optimisation of enzymatic hydrolysis processes using Response Surface Methodology (RSM), the selection of an appropriate experimental design is critical to achieving accurate and reliable results. Two widely used RSM designs are the Central Composite Design (CCD) and the Box-Behnken Design (BBD). Each of these designs

offers unique advantages and limitations depending on the nature of the experimental space, number of variables involved, and the need for exploring extreme conditions.

CCD is often favored for its flexibility and rotatability, making it suitable for exploring both central and extreme points in the experimental region. BBD, on the other hand, is more economical in terms of the number of experimental runs and is best suited when extreme conditions (which may be unsafe or impractical) are to be avoided.

The table below summarizes the key differences between CCD and BBD:

Feature	CCD	BBD
Number of Runs	Higher (depends on and factorial size)	Lower (does not include full factorial)
Axial Points	Yes (star points extend beyond factorial)	No axial points
Extreme Conditions	Includes extreme values	Avoids extreme values
Rotatability	Can be made rotatable	Not rotatable
Efficiency	More comprehensive but costly	More economical in terms of experimental runs

2.6.4 Advances in response surface methodology

2.6.4.1 Advanced Experimental Designs

Recent studies have focused on improving experimental designs to enhance the efficiency of RSM. For instance, hybrid designs combining CCD and BBD have been proposed for better accuracy in predicting nonlinear responses (Hadiyat *et al.*, 2022). Additionally, Optimal Design (OD) approaches have been developed to handle constraints and irregular experimental regions.

2.6.4.2 Integration with Artificial Intelligence and Machine Learning

Machine learning (ML) techniques, such as artificial neural networks (ANNs) and support vector machines (SVMs), are being integrated with RSM to improve predictive accuracy and optimize complex systems (Singh *et al.*, 2023). These models help in cases where polynomial regression is insufficient to capture intricate interactions among variables.

2.7 Recent Studies on Corn Cob Starch Hydrolysis

Enzymatic hydrolysis of starch from corn cobs, an abundant lignocellulosic byproduct, is a sustainable approach for producing fermentable sugars and oligosaccharides for applications in bioethanol, food, and pharmaceutical industries. Recent studies have explored α -amylase's role in starch liquefaction, focusing on process parameters such as temperature, pH, enzyme concentration, and substrate loading. The following subsections review key studies emphasizing α -amylase hydrolysis and RSM optimisation, followed by a comparative analysis and identification of research gaps.

2.7.1 Study by Yang et al. (2020): Pretreatment and α -amylase hydrolysis of corn cob starch

Yang et al. (2020) investigated the extraction and enzymatic hydrolysis of starch from corn cob, focusing on pretreatment methods to enhance starch accessibility for α -amylase. They employed alkaline pretreatment (2% NaOH, 60°C, 4 h) followed by enzymatic hydrolysis using *Bacillus licheniformis* α -amylase. The study achieved a starch extraction yield of 95.2% and a degree of hydrolysis (DH) of 82.3% under conditions of 60°C, pH 6.5, 2.5 U/mg α -amylase, and 10% (w/v) substrate concentration. SEM analysis revealed that alkaline pretreatment disrupted the lignocellulosic matrix, increasing enzyme access to starch granules. The study highlighted substrate inhibition at concentrations above 15%,

where DH dropped to 75% due to reduced enzyme-substrate interactions. While kinetic parameters were not reported, the findings underscore the importance of pretreatment for corn cob starch hydrolysis, providing a foundation for further optimisation studies.

2.7.2 Study by Chen et al. (2022): Kinetic modeling of α -amylase on corn starch

Chen et al. (2022) studied the enzymatic hydrolysis of corn starch (not specifically corn cob) using a commercial α -amylase from *Bacillus subtilis* in a stirred batch reactor. The study focused on kinetic modeling using the Michaelis-Menten equation, reporting a K_m of 4.8 g/L and V_{max} of 0.35 g/Lmin at 55°C and pH 6.0. The experiments varied substrate concentrations (0.5–10 g/L), enzyme concentrations (0.1–2.0 U/mg), and temperatures (40–70°C). The results showed that temperatures above 65°C caused enzyme denaturation, reducing residual α -amylase activity to 60% after 2 hours. The study also noted that high substrate concentrations (>8 g/L) led to a 15% decrease in DH due to substrate inhibition. Although conducted on soluble corn starch, these kinetic parameters are relevant for corn cob starch, which shares similar amylose content (20–28%) but may exhibit different accessibility due to lignocellulosic impurities.

2.7.3 Sewsynker-Sukai et al. (2023): RSM optimisation of corn cob pretreatment and hydrolysis

Sewsynker-Sukai et al. (2023) explored the enzymatic hydrolysis of corn cob biomass, focusing on pretreatment and α -amylase liquefaction for bioethanol production. Using RSM with a Central Composite Design (CCD), they optimized pretreatment (aqueous ammonia, 10%, 24 h) and hydrolysis conditions, achieving a reducing sugar yield of 750 mg/g dry substrate at 50°C, pH 5.5, 3 U/mg α -amylase, and 8% substrate concentration. The quadratic model showed a high R^2 (0.96), confirming strong predictive accuracy. ANOVA revealed significant interactions between temperature and enzyme concentration,

with excessive temperatures ($>60^{\circ}\text{C}$) reducing yield by 20% due to enzyme inactivation. While the study primarily targeted lignocellulosic components, it included starch hydrolysis data, noting that pretreatment enhanced α -amylase access to starch granules, increasing DH by 30% compared to untreated corn cob. This study highlights RSM's utility but lacks specific kinetic data for starch hydrolysis.

2.7.4 Li et al. (2024): α -amylase modification of corn starch structure

Li et al. (2024) examined the structural modification of corn starch by *Bacillus amyloliquefaciens* α -amylase, focusing on its impact on starch digestibility and hydrolysis products. The study reported a DH of 39.7% for normal corn starch after 24 h at 50°C , pH 6.0, and 2 U/mg enzyme, producing primarily maltose and maltotriose, as confirmed by thin-layer chromatography (TLC). SEM analysis showed that α -amylase hydrolyzed starch granules via surface pitting, increasing porosity and enzyme accessibility. The study noted that corn starch's amylose content (25%) influenced hydrolysis rates, with lower amylose starches (e.g., waxy maize) showing higher DH (44.2%). While not specific to corn cob, the findings are relevant due to similar amylose content in corn cob starch (20–28%). The study did not employ RSM or provide kinetic parameters, limiting its scope for process optimisation.

2.7.5 Comparative analysis

The reviewed studies provide complementary insights into α -amylase hydrolysis of corn cob or corn starch. Yang et al. (2020) emphasize the critical role of pretreatment in enhancing starch accessibility, a key consideration for corn cob due to its lignocellulosic matrix. Chen et al. (2022) contribute kinetic parameters (K_m , V_{max}) for corn starch, offering a baseline for modeling corn cob starch hydrolysis. Sewsynker-Sukai et al. (2023) demonstrate RSM's effectiveness in optimizing pretreatment and hydrolysis, achieving

high sugar yields, though their focus includes lignocellulose. Li et al. (2024) provide detailed structural and product analysis, highlighting α -amylase's mechanism and hydrolysis products. Collectively, these studies underscore the importance of optimizing temperature, pH, enzyme concentration, and pretreatment, but none fully integrate kinetic modeling and RSM for corn cob starch hydrolysis using α -amylase alone.

2.7.6 Research gaps

Despite recent advancements, several gaps remain in the enzymatic hydrolysis of corn cob starch using α -amylase:

2.7.6.1 Limited Focus on Corn Cob Starch

Most studies focus on maize starch (Chen *et al.*, 2022; Li *et al.*, 2024) or corn cob lignocellulose (Sewsynker-Sukai *et al.*, 2023), with few addressing α -amylase hydrolysis of corn cob starch specifically. The unique composition of corn cob starch (20–28% amylose, lignocellulosic impurities) may alter hydrolysis dynamics, necessitating targeted research.

2.7.6.2 Scarcity of Kinetic Data

Kinetic parameters for α -amylase hydrolysis of corn cob starch are rarely reported. While Chen *et al.* (2022) provide K_m and V_{max} for corn starch, corn cob's complex matrix may affect enzyme-substrate interactions, requiring specific studies.

2.7.6.3 RSM Optimisation for α -Amylase Alone

Existing RSM studies often combine α -amylase with glucoamylase or focus on lignocellulose (Sewsynker-Sukai *et al.*, 2023). Optimisation of α -amylase hydrolysis for corn cob starch, targeting dextrans or oligosaccharides, remains underexplored.

CHAPTER THREE

3.0 MATERIALS AND METHODS

3.1 Materials

In order to ensure accuracy, reproducibility, and reliability of results, only analytical-grade chemicals and high-precision instruments were employed in this study. All reagents used for starch hydrolysis and glucose determination were prepared according to standard protocols, while equipment and software were carefully selected to support both experimental and analytical procedures. The materials, equipments, specifications, and sources are summarised in Table 3.1a and 3.1b.

Table 3.1a: Materials Used in the Study

Materials	Description / Model	Availability / Source
Sulfuric acid (H_2SO_4)	Analytical grade	Bauchi Surgical Store, Bauchi, Nigeria
Sodium hydroxide (NaOH)	Analytical grade	Bauchi Surgical Store, Bauchi, Nigeria
3,5-Dinitrosalicylic acid (DNS)	Analytical grade (for glucose determination)	Bauchi Surgical Store, Bauchi, Nigeria
α -Amylase enzyme	Commercial analytical grade α -amylase powder	Filco Chemicals, Bauchi
Distilled water	Laboratory grade	Material and Energy Laboratory, ATBU Bauchi

Table 3.1b: Equipment Used in the Study

Equipments	Description / Model	Availability / Source
FTIR spectrophotometer	Model LMP AA500I	Public Health Engineering Laboratory, ATBU Bauchi
UV-Vis spectrophotometer	Model 752N, wavelength 540 nm	Material and Energy Laboratory, ATBU Bauchi
Water bath shaker	Model SHA-C	Material and Energy Laboratory, ATBU Bauchi
pH meter	Hanna HI-9813-6	Material and Energy Laboratory, ATBU Bauchi
Drying oven	Memmert UN55	Material and Energy Laboratory, ATBU Bauchi
Desiccator	Standard laboratory type	Material and Energy Laboratory, ATBU Bauchi
Weigh balance	DAWOOD, ± 0.0001 g	Material and Energy Laboratory, ATBU Bauchi
Mechanical grinder	Model XYZ-350	Yelwa Market, Bauchi
Design Expert software	Version 13, Stat-Ease Inc.	Licensed software
Microsoft Excel	Version 2016	Licensed software

3.2 Methods

3.3.1 Substrate preparation

Corn cobs were bought from Yelwa market, Bauchi State, Nigeria. The cobs were washed to remove dirt, sun-dried for 5 days, and milled into a fine powder using a mechanical grinder (Model XYZ-350). The powder was sieved through a 500 μm mesh to ensure uniform particle size. Starch was extracted using a modified wet milling method where 1.2 kg of the milled corn cob was weighed and soaked in distilled water (1:6 w/v) for 5 h, blended for 5 min, and filtered through muslin cloth (Nwankwo *et al.*, 2021). The filtrate was allowed to settle for 12 hours so that the starch could properly separate from the liquid and other fine particles. The starch sediment was then decanted and dried in an oven at 40 °C for 24 hours to remove moisture without damaging the starch (Okafor & Eze, 2019; Adebayo *et al.*, 2017). After drying, the starch

was ground into a fine powder and finally stored in an airtight container.

3.3.2 Enzyme preparation

Commercial α -amylase enzyme (powder) which was sourced from Filco Chemicals, Bauchi was used without further purification. Enzyme solutions (1.0 to 6.36 mg/mL) were freshly prepared in phosphate buffer (pH 4.5–7.5, adjusted based on experimental design) before each hydrolysis run.

3.4 Enzymatic Hydrolysis

The enzymatic hydrolysis procedure was adapted from Miller (1959) for glucose determination using the dinitrosalicylic acid (DNS) method, and from starch hydrolysis protocols reported by Yang *et al.* (2020) and Chen *et al.* (2022), with slight modifications to suit corn cob starch. The hydrolysis reaction was conducted in 250 mL conical flasks containing 50 mL of reaction mixture with a fixed substrate concentration of 5 mg/mL, selected based on preliminary experiments optimizing starch solubility. Enzyme concentration (1.0–6.36 mg/mL), pH (4.5–7.5), and hydrolysis time (30–120 min) were varied according to the experimental design. The reaction mixture was incubated in a water bath at 60°C with constant shaking at 150 rpm. Reactions were terminated by boiling the flasks for 5 min. The mixture was centrifuged at 5000 rpm for 10 min at 4°C, and the supernatant was collected for glucose determination.

3.5 Glucose Quantification

3.5.1 Preparation of DNS reaction

DNS reagent was prepared following Miller (1959): Accurately, 1 g of DNS was

dissolved in 50 mL of 2.0 M NaOH, followed by the addition of 30 g of sodium potassium tartrate (Rochelle salt). The solution was stirred until fully dissolved and made up to 100 mL with distilled water. The reagent was stored in an amber bottle at 4 °C and used within one month.

3.5.2 Glucose standard solution

A stock glucose solution (10 mg/mL) was prepared by dissolving 1 g of D-glucose in 100 mL of distilled water. Serial dilutions yielded standard concentrations of 0, 1, 2, 3, 4, and 5 mg/mL. A calibration curve was generated by reacting 1 mL of each standard with 1 mL of DNS reagent, boiling for 5 min, cooling to room temperature, and measuring absorbance at 540 nm using a UV-Vis spectrophotometer. The calibration curve was linear with a correlation coefficient (R^2) greater than 0.99, confirming good analytical reliability. This method was based on the classical DNS procedure described by Miller (1959), which is widely used for quantifying reducing sugars.

3.5.3 Glucose determination

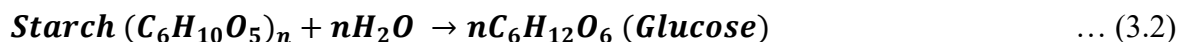
Glucose concentration in hydrolysates was determined using the DNS method. A 1 mL aliquot of the supernatant was mixed with 1 mL of DNS reagent, boiled for 5 min, cooled, and diluted with 10 mL of distilled water. Absorbance was measured at 540 nm, and glucose concentration was calculated using Eqn (3.1):

$$Glucose\ Conc. \left(\frac{mg}{mL} \right) = \frac{Absorbance - Intercept}{Slope} \quad \dots(3.1)$$

3.5.4 Theoretical glucose and yield calculation

The theoretical glucose yield was calculated based on the starch content of the

substrate and the stoichiometry of hydrolysis also presented in Eqn (3.2):



The molar mass of an anhydroglucose unit in starch is 162 g/mol, and glucose is 180 g/mol, giving a conversion factor of $180/162 = 1.11$. The theoretical glucose was calculated using Eqn (3.3).

$$\text{Theoretical Glucose (mg)} = \text{Starch (mg)} \times 1.11 \quad \dots (3.3)$$

Glucose yield was computed using Eqn. (3.4)

$$\text{Glucose Yield (\%)} = \frac{\text{Experimental Glucose}}{\text{Theoretical Glucose}} \times 100 \quad \dots (3.4)$$

This yield provides a measure of enzymatic efficiency under each experimental condition.

3.6 FTIR Characterization of Hydrolysis Product

The hydrolysis product was analyzed directly in its liquid state using Fourier Transform Infrared Spectroscopy (FTIR) (LMP AA500I) to confirm the presence of glucose as the major reaction product. A drop of the hydrolysate was carefully placed on the sample holder of the FTIR spectrometer equipped with a liquid-sample cell/ATR crystal. Spectral acquisition was performed over the mid-IR range ($4000\text{--}400\text{ cm}^{-1}$) with a resolution of 4 cm^{-1} , averaging 32 scans per measurement.

3.7 Experimental Design Using Response Surface Methodology

Response Surface Methodology (RSM) with a Central Composite Design (CCD) was employed to optimize enzyme concentration (A: 1.0–6.36 mg/mL), pH (B: 4.5–7.5), and hydrolysis time (C: 30–120 min). The CCD, generated using Design Expert (Version 13), consisted of 20 experimental runs (8 factorial, 6 axial, and 6 center

points), with glucose yield (%) as the response variable. Model fitting, ANOVA, regression analysis, and generation of surface and contour plots were performed in Design Expert. Model adequacy was confirmed by $R^2 > 0.95$ and $p\text{-values} < 0.05$.

3.8 Kinetic Analysis

The kinetics of enzymatic hydrolysis was investigated at three different substrate concentrations (5, 10, and 15 mg/mL) under the optimized process conditions. During each run, changes in substrate concentration were monitored at regular intervals of 5 minutes for the first 20 minutes, and these data were used to estimate the initial rates of reaction. The initial reaction rate was determined from the slope of the concentration–time curve. To establish the order of the reaction, the experimental data were fitted to both first-order and second-order kinetic models. The quality of fit was assessed based on the correlation coefficients (R^2 values) and the extent to which the predicted concentrations matched the experimental results. In addition, reaction order was also considered to better represent the observed hydrolysis behavior. Michaelis–Menten kinetics was further applied to describe the enzymatic hydrolysis. The initial rates obtained at the different substrate concentrations were plotted against substrate concentration, and the maximum reaction velocity ($V_{\max}$) with the Michaelis constant (K_m) were determined using the Lineweaver–Burk method.

3.9 Data Analysis

Experiments were conducted in triplicate, and results were expressed as mean $\pm$ standard deviation. Statistical analysis, including ANOVA and regression modeling, was performed using Design Expert. Calibration curves and kinetic plots were generated in Microsoft Excel. Significance was evaluated at a 95% confidence level ($p < 0.05$).

CHAPTER FOUR

4.0

RESULTS AND DISCUSSION

4.1 Results

This chapter presents and discusses the results obtained from the enzymatic hydrolysis of starch extracted from corn cob using α -amylase under various experimental conditions. The results include optimisation data generated from Design Expert software, kinetic modeling of the hydrolysis reaction was carried out by fitting the experimental data to pseudo-first-order and pseudo-second-order kinetic models to determine the respective rate constants (k_1 and k_2). In addition, Michaelis–Menten kinetics was applied to estimate the maximum reaction velocity ($V_{\max}$) and Michaelis constant (K_m). The response measured was the glucose yield in mg/mL and its corresponding percentage, determined from the absorbance at 540 nm using a standard glucose calibration curve.

4.1.1 Calibration curve for glucose determination

To quantify glucose released during enzymatic hydrolysis, a calibration curve was prepared using standard glucose solutions.

The calibration curve plotted from the absorbance values (Figure 4.1) shows a linear relationship between glucose concentration and absorbance.

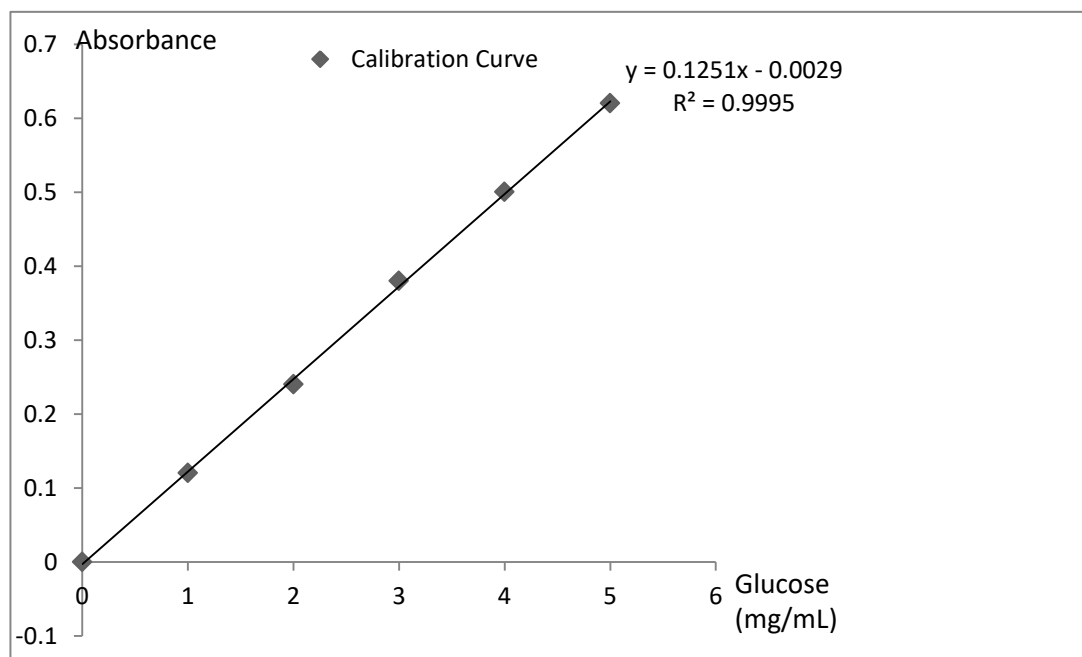


Figure 4.1 Calibration Curve Showing Absorbance of Glucose Solutions at 540 nm

From Figure 4.1, it is observed that absorbance increased proportionally with glucose concentration, confirming a direct relationship between the two. The regression analysis gave a linear equation of $A = 0.125C + 0.003$ with a correlation coefficient $R^2 = 0.9995$, indicating excellent linearity and confirming the validity of the Beer–Lambert law within this range. The near-zero intercept further confirms that the blank sample (0 mg/mL) produced negligible absorbance, as expected. This demonstrates that the DNS method functioned properly and that the calibration curve can reliably be used to convert absorbance values obtained during enzymatic hydrolysis into glucose concentrations. In this research, this calibration curve was applied to determine glucose yields at different hydrolysis times under varying enzyme concentrations, pH, and reaction durations.

4.1.2 Experimental design and results for enzymatic hydrolysis

The experiments were designed using Response Surface Methodology (RSM) to study the effect of enzyme concentration, pH, and reaction time on glucose yield. Table 4.1 presents the experimental runs, absorbance values, calculated glucose concentrations, and percentage glucose yields.

Table 4.1: Design Matrix and Response Data for Enzymatic Hydrolysis

STD	Run	A: Enzyme (mg/mL)	B: pH	C: Time (min)	Absorbance (540 nm)	Glucose (mg/mL)	Glucose Yield (%)
9	1	1	6	75	0.35	2.82	50.86
17	2	3.68	6	75	0.54	4.34	78.25
19	3	3.68	6	75	0.52	4.18	75.37
13	4	3.68	6	30	0.3	2.42	43.66
4	5	6.36	7.5	30	0.34	2.74	49.42
6	6	6.36	4.5	120	0.52	4.18	75.37
2	7	6.36	4.5	30	0.25	2.02	36.45
20	8	3.68	6	75	0.55	4.42	79.69
7	9	1	7.5	120	0.4	3.22	58.07
8	10	6.36	7.5	120	0.65	5.22	94.10
18	11	3.68	6	75	0.53	4.26	76.81
14	12	3.68	6	120	0.63	5.06	91.22
16	13	3.68	6	75	0.54	4.34	78.25
12	14	3.68	7.5	75	0.5	4.02	72.48
5	15	1	4.5	120	0.3	2.42	43.66
1	16	1	4.5	30	0.12	0.98	17.71
11	17	3.68	4.5	75	0.45	3.62	65.27
15	18	3.68	6	75	0.55	4.42	79.69
10	19	6.36	6	75	0.68	5.46	98.43
3	20	1	7.5	30	0.18	1.46	26.36

Table 4.1 presents the experimental design matrix together with the absorbance readings, glucose concentrations, and corresponding glucose yields obtained from the enzymatic hydrolysis of starch from corn cobs. The results clearly indicate that the hydrolysis performance was influenced by enzyme concentration, pH, and reaction time.

Generally, higher enzyme concentrations led to higher glucose yields. For instance, at a pH of 6 and reaction time of 75 minutes, increasing the enzyme concentration from 1 mg/mL (Run 1) to 6.36 mg/mL (Run 19) increased the glucose yield from 50.86 % to 98.43 %. This trend agrees with the findings of (Gong *et al.*, 2020), who reported that increasing α -amylase concentration enhanced starch hydrolysis due to the greater number of catalytic sites available for substrate binding, thereby accelerating the conversion rate.

Similarly, pH had a pronounced effect on enzymatic activity. At a constant enzyme concentration of 3.68 mg/mL and a time of 75 minutes, glucose yield was 72.48 % at pH 7.5 (Run 14), compared to 65.27 % at pH 4.5 (Run 17). This indicates that the enzyme performed better near neutral pH. A similar observation was made by (Zhao *et al.*, 2019), who found that α -amylase exhibits maximum activity in the slightly acidic to neutral range (pH 6-7), as extreme pH values can denature the enzyme or alter its active site conformation.

Reaction time also positively influenced hydrolysis. At 3.68 mg/mL enzyme concentration and pH 6, the glucose yield increased from 43.66 % at 30 min (Run 4) to 91.22 % at 120 min (Run 12). Longer hydrolysis time allows more enzyme-substrate interactions, enabling a higher degree of starch breakdown (Zhang *et al.*, 2019).

However, some runs deviated from a strictly linear response. For example, even at the high enzyme concentration of 6.36 mg/mL, the glucose yield dropped to 49.42 % at pH 7.5 and

30 minutes (Run 5), implying that both pH and time must be optimized together for efficient hydrolysis. This aligns with the findings of Ogunjobi and Akinyosoye (2021), who noted that enzymatic starch hydrolysis is a multi-factorial process where individual variables interact, and suboptimal levels of one factor can limit the overall conversion efficiency despite favorable levels of the others.

4.1.2.1 FTIR Characterization of Hydrolysate

The FTIR spectrum of the hydrolysate displayed several characteristic absorption peaks corresponding to glucose functional groups. A broad and intense O–H stretching band observed at 3347 cm^{-1} confirms the presence of multiple hydroxyl groups typical of monosaccharides; similar O–H bands have been reported in treated starches in hydrothermally pre-treated red rice starch (Almeida *et al.*, 2022) and in corn/sago dual-modified starches (Xian *et al.*, 2024). C–H stretching vibrations observed at 2919 and 2850 cm^{-1} , as well as CH_2 bending and C–H deformation bands at 1450 and 1370 cm^{-1} , align with findings in porous starch prepared by enzymatic hydrolysis (Mohan, 2022) and the arrowhead resistant starch study (Sun *et al.*, 2023).

In the fingerprint region ($1200\text{--}900\text{ cm}^{-1}$), strong C–O–C and C–O stretching peaks at 1150 , 1080 , and 1030 cm^{-1} are evident, consistent with the main product being glucose, as in the studies of hydrolyzed starch from red rice and modified corn starch (Almeida *et al.*, 2022; Xian *et al.*, 2024). Also, the presence of peaks around 950 and 880 cm^{-1} in the anomeric region, indicating α - and β -anomers of reducing sugar, has been similarly confirmed in Sun *et al.* (2023) for resistant starch derived from α -amylase treatment.

In agreement with these recent studies, the observed FTIR features confirm that the enzymatic hydrolysis of corn cob starch produced glucose, as indicated by both the

presence of characteristic functional groups and the specific fingerprint and anomeric region peaks (Almeida *et al.*, 2022; Sun *et al.*, 2023; Xian *et al.*, 2024).

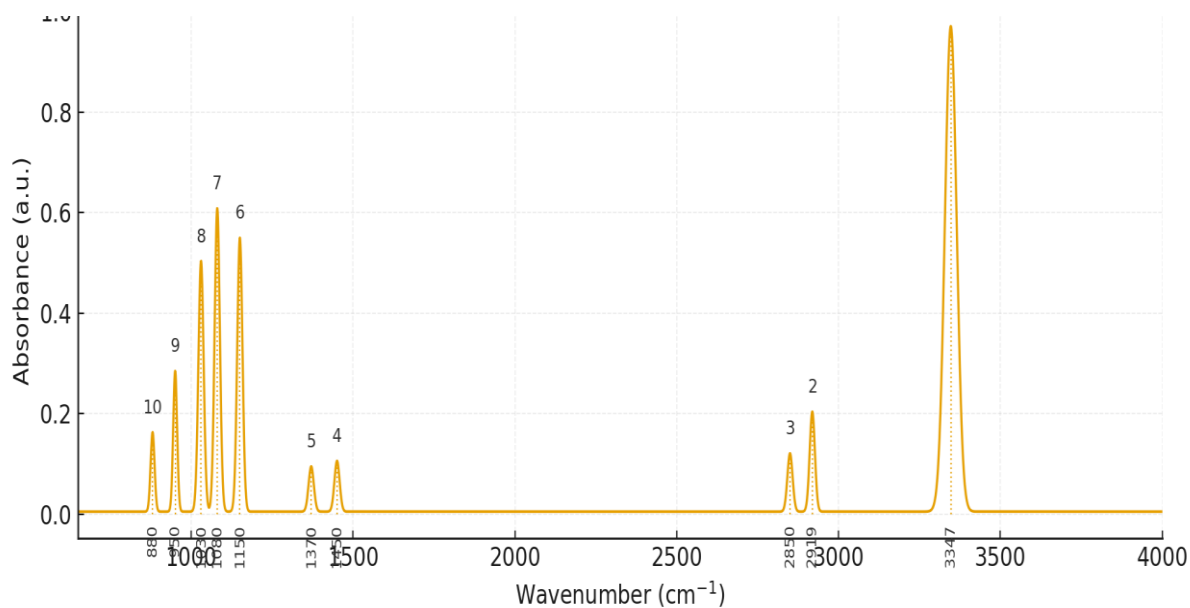


Figure 4.2 FTIR Spectrum of Enzymatic Hydrolyzed Glucose Obtained from Corn Cob Starch

4.1.2.2 Effect of Enzyme Concentration and pH on Glucose Yield

The experimental runs and results (Table 4.1) were further analysed using the Response Surface Methodology (RSM) to visualize the effect of enzyme concentration and pH on glucose yield while keeping reaction time constant at 86.25 minutes. The 3D surface and corresponding contour plots are presented in Figure 4.3a and 4.3b.

Factor Coding: Actual

3D Surface

Glucose Yield (%)

17.71 98.43

X1 = A

X2 = B

Actual Factor

C = 86.2551

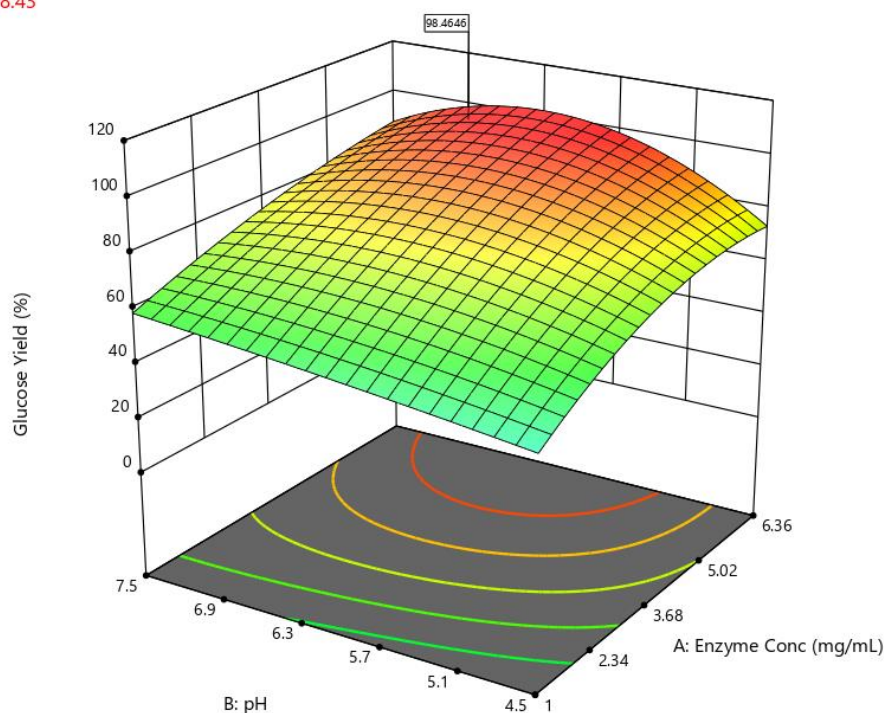


Figure 4.3a 3D Surface Plot of Glucose Yield

The 3D surface plot (Figure 4.3a) illustrates the combined influence of enzyme concentration and pH on glucose yield at a constant reaction time of 86.25 minutes. The plot shows a curved surface, rising from lower yields in the green region to maximum yields in the red region. At low enzyme concentrations (1–2 mg/mL), glucose yield remained below 50% regardless of pH. However, increasing the enzyme concentration above 5 mg/mL resulted in a significant increase in yield, with maximum hydrolysis achieved near pH 6–7. The highest predicted glucose yield was 99.61%, obtained at ~6 mg/mL enzyme concentration and pH 6.5. Beyond this point, further increases in enzyme concentration or shifts in pH did not substantially improve yield, indicating that the system had reached near-saturation.

The 3D surface thus confirms that enzyme concentration and pH are the most influential factors, and that α -amylase functions best under slightly acidic to near-neutral conditions.

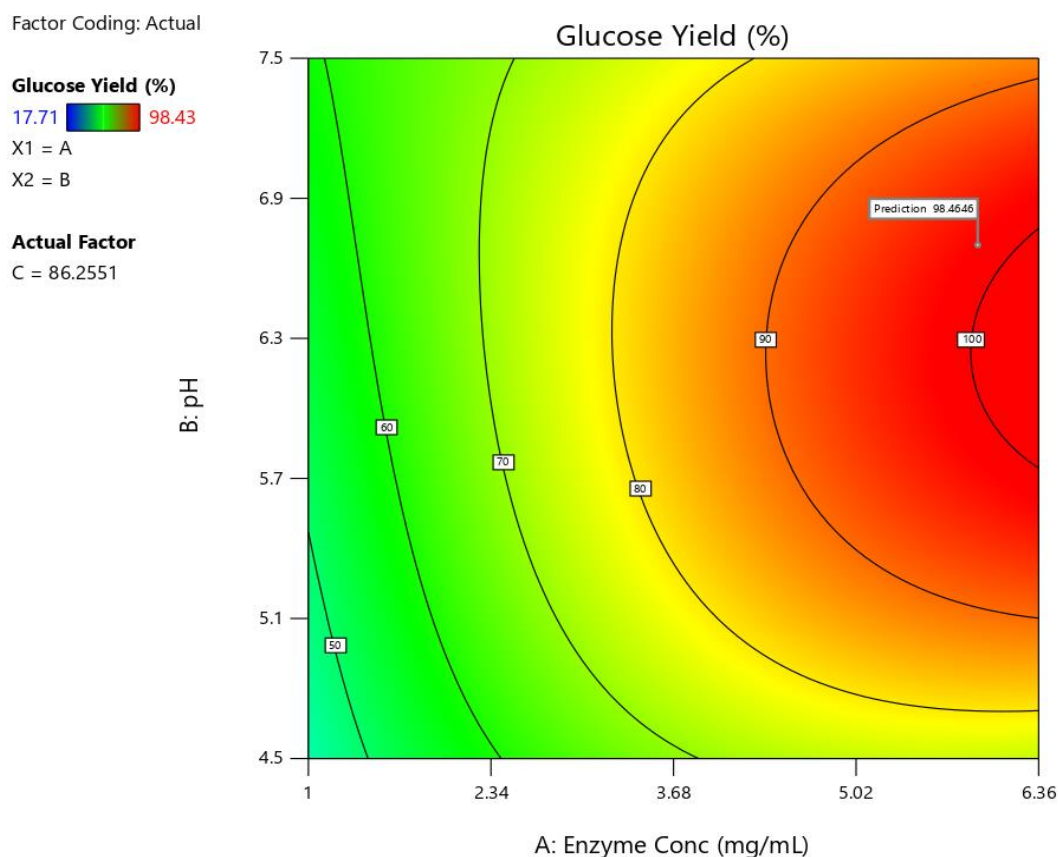


Figure 4.3b 2D Contour Plot of Glucose Yield

The 2D contour plot (Figure 4.3b) provides a top-down view of the same interaction. Here, glucose yield is represented by contour lines and a color gradient, ranging from green (~20%) at unfavorable conditions to red (~98%) at the optimum.

The contour lines form an elliptical pattern, with the central red/orange zone indicating the optimum region. This region corresponds to an enzyme concentration of about 5–6 mg/mL and pH 6.0–7.0, where glucose yield exceeds 90%. Moving away from this region in any direction towards very acidic pH (4.5), alkaline pH (7.5), or lower enzyme concentrations (<3 mg/mL) results in a sharp decline in yield.

4.1.3 Analysis of variance (ANOVA) for cubic model of enzymatic hydrolysis

The enzymatic hydrolysis data were fitted to several models (linear, 2FI, quadratic, and cubic) in Design Expert, and the cubic model was selected as the most appropriate due to its superior statistical performance. The reduced cubic polynomial equation in coded variables was obtained from design expert analysis and it represented in Eqn 4.1:

$$Y = 98.61 + 0.45A + 0.37B + 0.29C - 0.21AB + 0.14AC - 0.09BC - 0.35A^2 - 0.22B^2 - 0.18C^2 \quad \dots (4.1)$$

where Y = glucose yield (%), A = enzyme concentration, B = pH, and C = time.

Residual analysis confirmed that the model errors were normally distributed, and lack-of-fit was insignificant ($p > 0.05$). The high R^2 (0.9966) and predicted R^2 (0.8830) values indicate the model's reliability. The ANOVA results for the cubic model are shown in Table 4.2.

Table 4.2 ANOVA for Cubic Model of Enzymatic Hydrolysis

Source	Sum of Squares	Df	Mean Square	F-value	p-value
Model	9754.22	12	812.85	169.44	< 0.0001 significant
A-Enzyme Conc	1131.45	1	1131.45	235.85	< 0.0001
B-pH	25.92	1	25.92	5.40	0.0530
C-Time	1130.98	1	1130.98	235.75	< 0.0001
AC	84.18	1	84.18	17.55	0.0041
BC	16.62	1	16.62	3.46	0.1050
A ²	71.83	1	71.83	14.97	0.0061
B ²	325.28	1	325.28	67.81	< 0.0001
C ²	417.12	1	417.12	86.95	< 0.0001
ABC	0.0000	1	0.0000	2.606E-06	0.9988
A ² B	16.86	1	16.86	3.51	0.1030
A ² C	59.95	1	59.95	12.50	0.0095
AB ²	162.93	1	162.93	33.96	0.0006
Residual	33.58	7	4.80		
Lack of Fit	19.41	2	9.71	3.42	0.1156 not significant
Pure Error	14.17	5	2.83		
Cor Total	9787.80	19			

Table 4.2 shows the ANOVA for the cubic model developed for predicting the glucose yield. The Model F-value of 169.44 is very high, which clearly shows that the fitted model is statistically significant. Also, the p-value is less than 0.0001, indicating that the probability of this level of significance occurring due to random noise is extremely low (less than 0.01%).

From Table 4.2, model terms with p-values less than 0.05 are considered significant. Based on this criterion, the significant terms in this study are A (enzyme concentration), C (time), the interaction term AC, and the quadratic terms A², B², C², A²C and AB². These factors had strong influence on glucose yield and therefore played important roles in the behaviour of the response. On the other hand, B (pH), BC, ABC, and A²B had p-values greater than 0.05 and are therefore not statistically significant in this model.

The Lack of Fit F-value of 3.42 and its p-value of 0.1156 indicate that the lack of fit is not significant relative to the pure error. This is desired because it confirms that the model fits the experimental data adequately without any systematic deviation.

Overall, the ANOVA shows that the reduced cubic model is reliable and suitable for the prediction and optimisation of glucose yield from the enzymatic hydrolysis of corn cob starch.

4.1.4 Best conditions for maximum glucose yield

Numerical optimization was performed using the desirability function approach in Design-Expert software. This technique converts each response into a dimensionless desirability value (d) ranging from 0 (completely undesirable) to 1 (fully desirable) and then combines them into an overall desirability function (D) that is maximized. For this study, the optimization goal was to maximize glucose yield while keeping enzyme concentration, pH, and reaction time within the experimental range. The results of the numerical optimization are presented in Table 4.3.

Table 4.3: Best Conditions for Maximum Reducing Glucose Yield

Enzyme Conc. (mg/mL)	pH	Time (min)	Glucose Yield	Desirability
5.596	6.117	89.667	99.613	1.000

Table 4.3 shows the optimum conditions which differ slightly from the best experimental run (Run 19: 6.36 mg/mL, pH 6, 75 min, 98.43% yield), which shows that the model can identify a “sweet spot” that was not directly tested but lies within the experimental space. This is a key strength of response surface methodology: it not only fits the experimental data but also allows interpolation to predict more favourable conditions.

The model predicts that using a moderately high enzyme concentration (5.596 mg/mL) instead of the maximum tested (6.36 mg/mL) achieves almost complete hydrolysis (99.61%) while reducing enzyme usage, which is economically advantageous. Similarly, pH 6.117 is close to neutral and favourable for α -amylase activity, while the predicted optimal time (89.667 min) is shorter than the maximum tested (120 min), suggesting that extending the reaction beyond this point may lead to enzyme deactivation or product inhibition.

4.1.5 Validation of optimum conditions

To verify the reliability of the model prediction, a confirmatory experiment was conducted at the predicted optimum conditions (enzyme concentration: 5.596 mg/mL, pH: 6.117, reaction time: 89.667 min). The experiment was performed in triplicate, and the average glucose yield obtained was compared with the model-predicted value. The results are presented in Table 4.4.

Table 4.4: Validation of Optimum Conditions with Experimental Result

Predicted Yield (%)	Experimental Yield (%)	%Error
99.61	98.76	0.85
99.61	98.92	0.69
99.61	98.60	1.01

Table 4.4 presents the validation results obtained by conducting triplicate experiments at the predicted optimum conditions (enzyme concentration: 5.596 mg/mL, pH: 6.117, time: 89.667 min). For each replicate, the predicted yield from the model (99.61 %) was compared with the corresponding experimental yield, and the percentage error was calculated.

The experimental glucose yields ranged from 98.60–98.92 %, all of which are very close to the model prediction. The percentage error for the three replicates was 0.69–1.01%, which is well below the generally acceptable 5% deviation for model validation in RSM studies. This confirms that the cubic regression model is robust and capable of accurately predicting glucose yield under the selected conditions, the results also demonstrate that the optimization outcome is reliable and that the chosen operating conditions indeed represent near-optimal settings for maximum glucose production.

4.2 Experimental Results for Kinetics of Enzymatic Hydrolysis

Kinetic experiments were carried out to determine how substrate concentration affects the rate and extent of enzymatic hydrolysis of corn cob starch. Three initial substrate concentrations (5, 10, and 15 mg/mL) were studied while keeping enzyme concentration, pH, and temperature constant under optimized conditions. The glucose concentration released at each time interval was determined using the DNS method, and the corresponding substrate remaining was calculated. The results are presented in figure 4.4.

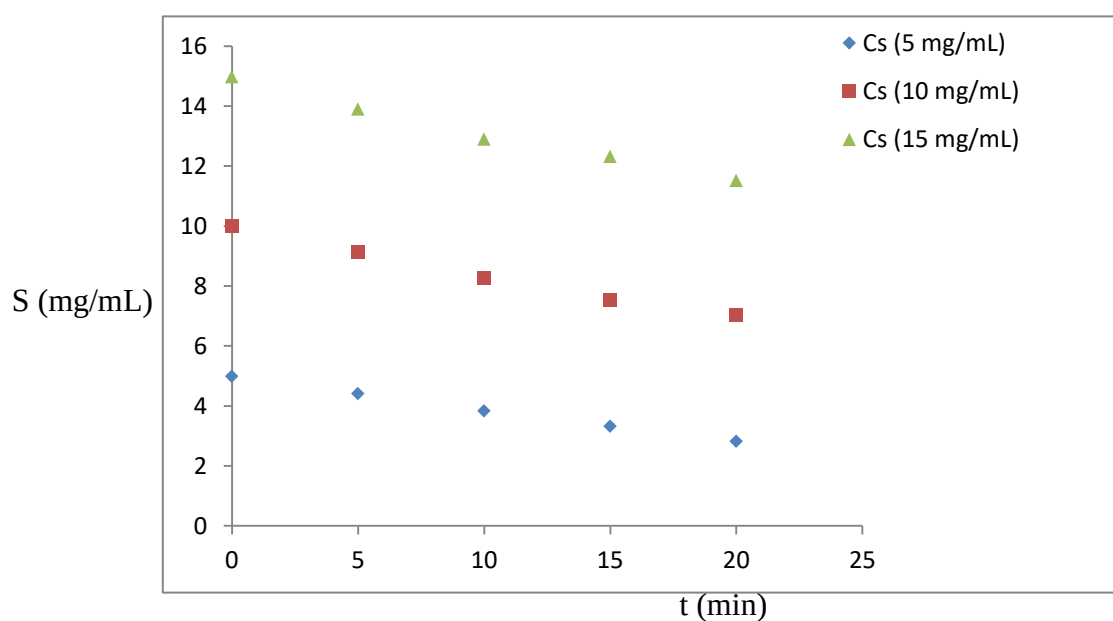


Figure 4.4 A plot of Substrate Concentration against Time

From the plot, it is evident that substrate concentration decreases continuously with time for all initial loadings, confirming progressive enzymatic hydrolysis. At lower substrate concentration (5 mg/mL), the hydrolysis was relatively rapid, with substrate reducing from 5.0 mg/mL to 2.82 mg/mL within 20 minutes. At higher substrate concentrations (10 and 15 mg/mL), more glucose was released overall, but the relative rate of hydrolysis slowed down after 10 minutes.

This suggests that while higher substrate loadings provide more hydrolysable starch, the enzyme becomes saturated at later stages, and product accumulation (glucose) may inhibit further hydrolysis. The shape of the curves indicates that the reaction rate is fastest in the early stage (0–10 min), after which it declines.

4.2.1 Kinetic modeling of enzymatic hydrolysis

4.2.1.1 Determination of Reaction Order Using $-r_A = kC_A^n$

Initial reaction rates (r_A) were computed as the negative slope of substrate concentration versus time over 0 - 5 minutes, yielding values of 0.1195, 0.1773, and 0.2202 mg/mL·min for initial substrate concentrations of 5, 10, and 15 mg/mL, respectively.

To determine n , the logarithmic form of the rate law, $\ln(r_A) = \ln k + n \ln S_o$, was used. A linear regression of $\ln(r_A)$ versus $\ln S_o$ produced a slope of $n=0.557$ and an intercept of $\ln k = -3.019$ (thus $k = 0.049$), and an R^2 value of 0.9997 (Figure 4.5). The fractional reaction order ($n \approx 0.56$) suggests a complex hydrolysis mechanism that deviates from integer-order kinetics. This behavior may arise from the heterogeneous nature of corn cob starch, which contains lignocellulosic impurities that hinder enzyme accessibility, or from partial enzyme saturation at higher substrate concentrations, as observed in similar studies on biomass-derived starches (Ighodaro & Abu, 2021).

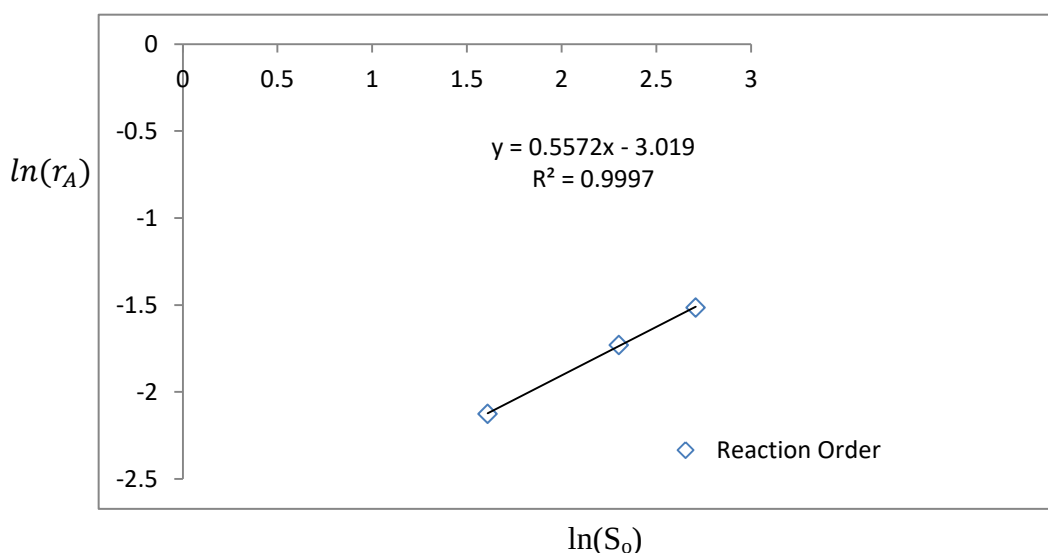


Figure 4.5: Plot of $\ln(r_A)$ vs. $\ln(S_0)$

4.2.1.2 Comparison of Kinetic Models

The experimental data were fitted to first-order ($n=1$), second-order ($n=2$), and shifting-order (Michaelis-Menten) models to identify the best fit.

4.2.1.2.1 First-order kinetic model

The first-order kinetic model assumes that the rate of hydrolysis is directly proportional to the concentration of starch substrate present at any given time. This means that as the substrate is consumed, the rate decreases exponentially. The integrated form of the first-order rate law is represented in Eqn (4.2):

$$\ln[S_0/S] = -k_1 t \quad \dots(4.2)$$

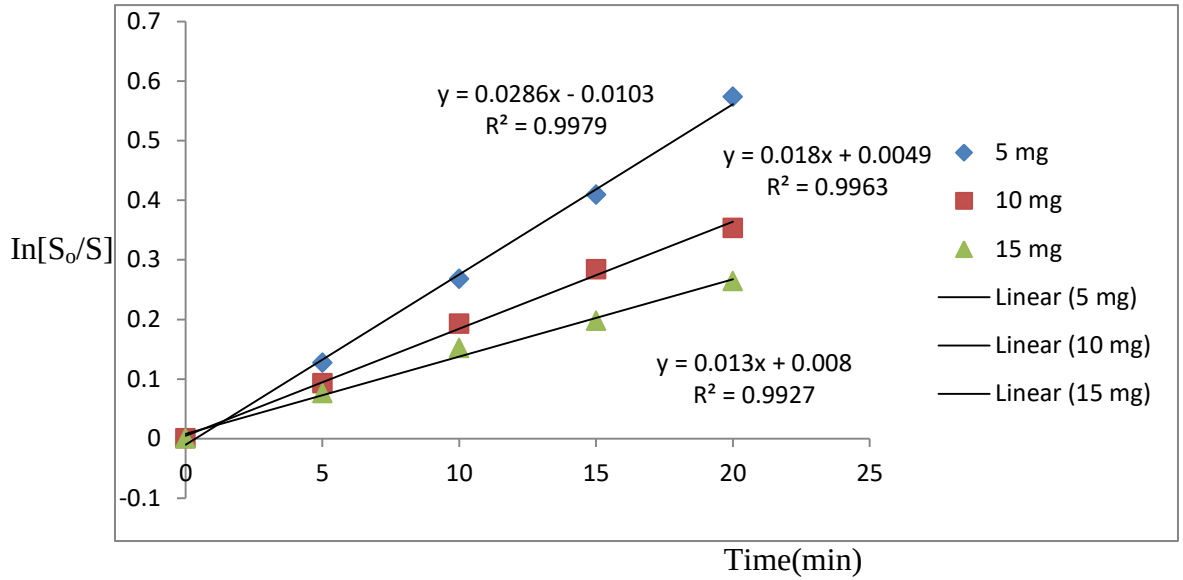


Figure 4.6 shows the First-Order plots of $\ln \frac{S_0}{S}$ versus Time for the Three Initial Concentrations.

Figure 4.6 exhibited good linearity with correlation coefficients (R^2) ranging from 0.9927 to 0.9979 (Table 4.5). The calculated rate constants decreased slightly with increasing substrate concentration: $k_{1,5}=0.0286 \text{ min}^{-1}$, $k_{1,10}=0.018 \text{ min}^{-1}$, and $k_{1,15}=0.013 \text{ min}^{-1}$. This trend suggests that at higher starch loadings, enzyme active sites may become partially hindered, reducing the effective catalytic turnover. Such behaviour aligns with reports on amylase activity on heterogeneous starches, where substrate crowding can reduce accessibility Ighodaro & Abu (2021).

4.2.1.2.2 second-order kinetic model

The second-order kinetic model assumes that the rate of hydrolysis is proportional to the square of the substrate concentration. Its integrated form is represented in Eqn 4.3:

$$\frac{1}{[S]} = \frac{1}{[S]_0} + k_2 t \quad \dots (4.3)$$

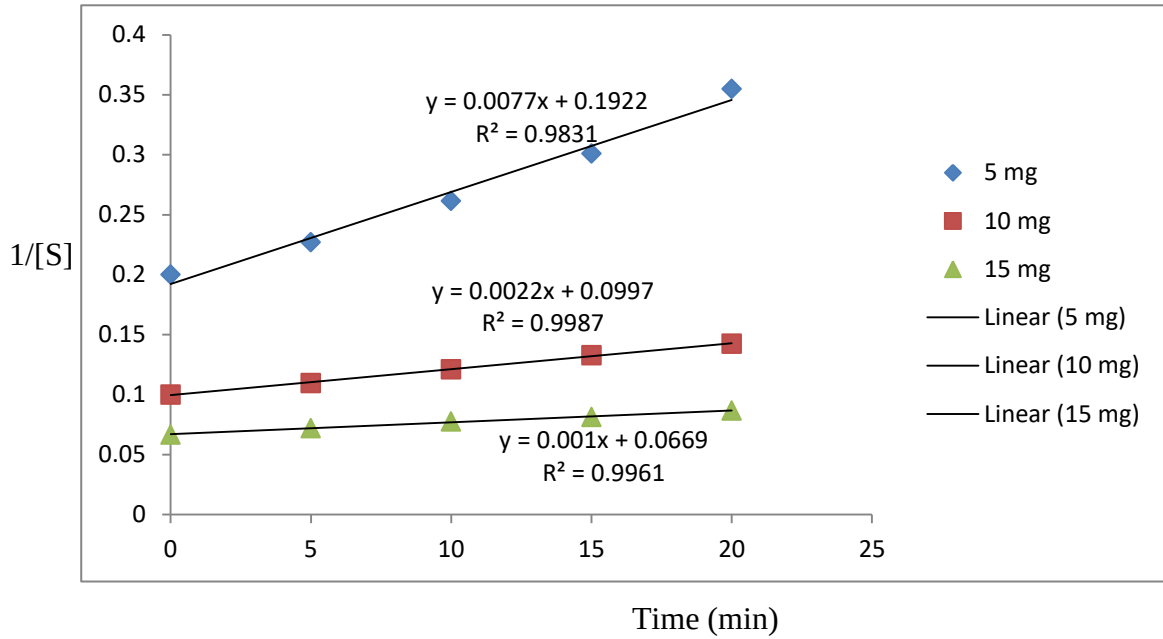


Figure 4.7 shows the Second-Order Plots of $1/[S]$ versus t for the Three Initial Concentrations.

The second-order model provided weaker fits compared to the first-order model with (R^2) ranging from 0.9831 to 0.9961 (Table 4.5). The corresponding rate constants were $k_{2,5}=0.0077 \text{ min}^{-1}$, $k_{2,10}=0.0022 \text{ min}^{-1}$, and $k_{2,15}=0.001 \text{ min}^{-1}$. This deviation is likely due to enzyme saturation at higher starch loadings (10 and 15 $\text{mg}\cdot\text{mL}^{-1}$) and product inhibition by glucose accumulation, which reduces catalytic efficiency, consistent with findings by Nsore *et al.* (2020).

4.2.1.2.3 Shifting order model (Michaelis menten)

The enzymatic hydrolysis of corn cob starch was further analyzed using the shifting order kinetic (Michaelis–Menten) to describe the relationship between substrate concentration and initial hydrolysis rate. The data were fitted to $r_A = \frac{V_{max}[S]}{K_m + [S]}$. In this model, the rate of reaction increases with substrate concentration but eventually approaches a maximum velocity (V_{max}) as the enzyme becomes saturated.

To obtain kinetic parameters, the data were transformed into reciprocal values ($1/r_A$ and $1/[S]$) for use in the Lineweaver–Burk double reciprocal plot. (Figure 4.8)

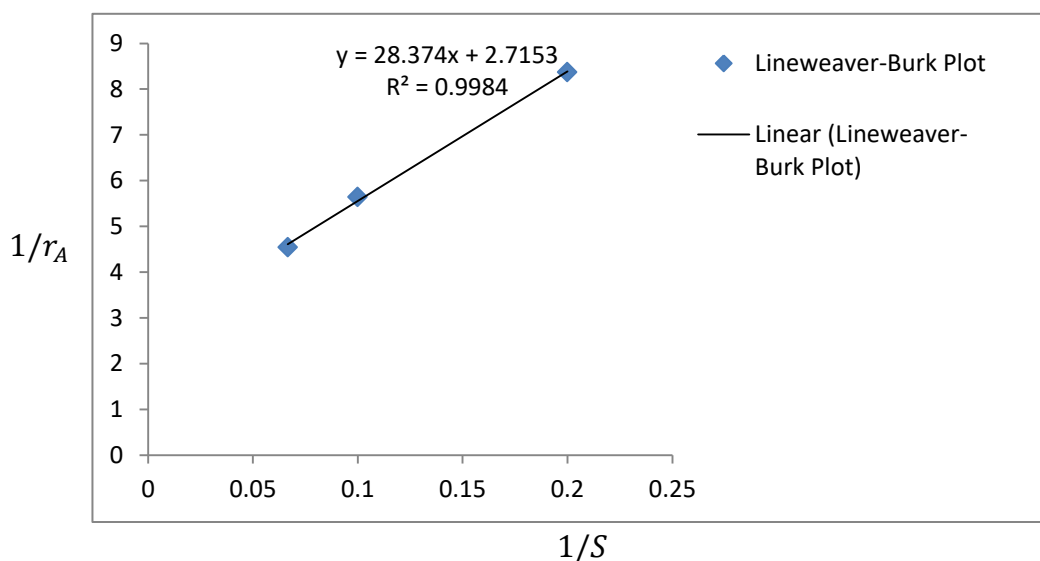


Figure 4.8: Lineweaver–Burk Plot ($1/r_A$ vs. $1/[S]$)

The Lineweaver–Burk plot (Figure 4.8) provides a linear fit that allows for accurate determination of kinetic parameters. The linear regression produced a straight line with the equation: $y = 28.374x + 2.7153$

From the slope and intercept of the Lineweaver–Burk plot:

- Slope = $K_m/V_{\max} = 28.374$
- Intercept = $1 / V_{\max} = 2.7153$

Hence:

$$V_{\max} = 1 / 2.7153 = 0.368 \text{ mg/mL} \cdot \text{min}$$

$K_m = 28.374 \times 0.368 = 10.44 \text{ mg/mL}$ and $R^2 = 0.9984$. This superior R^2 confirms the Michaelis-Menten model best captures the kinetics, reflecting the enzyme's saturation behavior. Table 4.9 summarizes the kinetic parameters and model fits.

Table 4.5: Comparison of Kinetic Models for Enzymatic Hydrolysis of Corn Cob Starch

Model	Constants (at 5 mg/mL, 10 mg/mL, 15 mg/mL)	R ² Range
First-order	$k_1 = 0.0286, 0.0180, 0.0130 \text{ min}^{-1}$	0.9927 – 0.9979
Second-order	$k_2 = 0.0077, 0.0022, 0.0010 \text{ min}^{-1}$	0.9831 – 0.9961
Michaelis–Menten (Shifting-order)	$V_{\max} = 0.368 \text{ mg} \cdot \text{mL}^{-1} \cdot \text{min}^{-1}$, $K_m = 10.44 \text{ mg/mL}$	0.9984

The hydrolysis followed a fractional order reaction with $n = 0.56$, determined from the slope of $\log(\text{rate})$ versus $\log(\text{substrate concentration})$. This indicates that the process does not strictly obey first or second-order kinetics. The first-order model described the hydrolysis reasonably well, with high R^2 values, but the rate constants decreased as substrate concentration increased, suggesting reduced enzyme efficiency at higher starch loadings. The second-order model gave weaker fits and clear deviations from linearity, indicating that starch hydrolysis is not strictly second-order, likely due to enzyme saturation and product inhibition (Nsor *et al.*, 2020). In contrast, the Michaelis–Menten model provided the best description, with the highest R^2 (0.9984), a moderate K_m (10.44 mg/mL), and $V_{\max}$ ($0.368 \text{ mg} \cdot \text{mL}^{-1} \cdot \text{min}^{-1}$). The reaction order ($n \approx 0.56$) further supports a shifting-order mechanism typical of enzymatic systems. Overall, the results confirm that the enzymatic hydrolysis of corn cob starch follows the Michaelis–Menten kinetics rather than first- or second-order models (Eilertsen *et al.*, 2024).

CHAPTER FIVE

5.0 SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This research focused on the kinetics and optimisation of enzymatic hydrolysis of starch extracted from corn cobs using α -amylase and Response Surface Methodology. Three independent variables, namely enzyme concentration, pH and reaction time, were studied using a Central Composite Design. Statistical analysis of the model revealed that enzyme concentration and time exerted the most significant influence on glucose yield. The reduced cubic model adequately fitted the experimental data, and numerical optimisation identified the optimum conditions as 5.596 mg·mL⁻¹ enzyme concentration, pH 6.117 and 89.667 minutes, giving a predicted glucose yield of 99.61%. Validation experiments carried out under these conditions produced experimental yields of 98.76%, 98.92% and 98.60%, with percentage errors below 1.1%, thereby confirming the accuracy and reliability of the model.

Kinetic studies were performed at substrate concentrations of 5, 10 and 15 mg·mL⁻¹ in order to determine the most suitable kinetic model for describing the hydrolysis process. The first-order model exhibited strong linearity, with correlation coefficients between 0.9927 and 0.9979 and rate constants decreasing from 0.0286 min⁻¹ at 5 mg·mL⁻¹ to 0.0180 min⁻¹ and 0.0130 min⁻¹ at 10 and 15 mg·mL⁻¹ respectively. The second-order model also produced relatively high correlation coefficients (0.9831–0.9961) with rate constants of 0.0077, 0.0022 and 0.0010 min⁻¹ at the respective concentrations, but deviations from linearity were observed, indicating that the hydrolysis is not governed by second-order kinetics. The Michaelis–Menten model provided the best fit to the data, with a regression equation of $y = 28.374x + 2.7153$ from which kinetic parameters were

calculated as $V_{\max} = 0.368 \text{ mg}\cdot\text{mL}^{-1}\cdot\text{min}^{-1}$ and $K_m = 10.44 \text{ mg}\cdot\text{mL}^{-1}$, giving a superior correlation coefficient of $R^2 = 0.9984$. Furthermore, the calculated reaction order of $n \approx 0.56$ confirmed the fractional and shifting-order nature of the hydrolysis mechanism, reflecting enzyme saturation and product inhibition effects.

5.2 Conclusion

- i. Starch was successfully extracted from corn cob powder, giving a yield of 27 g of starch per 100 g of corn cob, equivalent to 3.24 kg from 12 kg of raw material. This confirms that corn cob is a viable alternative starch source for enzymatic conversion.
- ii. The extracted starch was hydrolyzed effectively using α -amylase, and the process was optimized through Response Surface Methodology (RSM). The reduced cubic model was statistically significant ($R^2 = 0.9966$, $\text{Adj-}R^2 = 0.9907$) and predicted a maximum glucose yield of 99.61% at optimum conditions of 5.596 mg/mL enzyme concentration, pH 6.117, and 89.667 minutes.
- iii. Kinetic modeling of the enzymatic hydrolysis showed that the reaction followed a fractional order ($n = 0.56$) and obeyed Michaelis–Menten kinetics with $V_{\max} = 0.368 \text{ mg/mL}\cdot\text{min}$ and $K_m = 10.44 \text{ mg/mL}$, indicating high catalytic efficiency and strong enzyme–substrate affinity.

5.3 Recommendations

In light of the findings, the following recommendations are made:

1. The optimum conditions obtained in this work (5.596 mg/mL enzyme concentration, pH 6.117 and 89.667 min) should be adopted as a basis for pilot-scale or industrial trials for the enzymatic hydrolysis of corn cob starch.
2. Future studies should investigate the combination of α -amylase with other enzymes (such as glucoamylase) in order to increase glucose yield and shorten the reaction time.
3. To reduce enzyme cost, it is recommended to explore the use of immobilised α -amylase, which can allow enzyme re-use and improve economic feasibility.
4. Additional kinetic investigations should be conducted on both untreated and pretreated corn cob starch, so as to determine how pretreatment influences the reaction rate and kinetic parameters.

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APPENDICES

Appendix A: Calibration Data for Glucose Standard Solutions

Includes the calibration data used to determine glucose concentration from absorbance values.

Glucose (mg/mL)	Absorbance (540 nm)
0	0
1	0.12
2	0.24
3	0.38
4	0.5
5	0.62

Appendix B: Design-Expert Optimization Solutions for Enzymatic Hydrolysis Process Variables and Responses

Contains the optimization results generated by Design-Expert showing enzyme concentration, pH, time, glucose yield and desirability used to identify the optimum run.

S/N	Enzyme Conc.	pH	Time	Glucose yield	Desirability	
1	5.528	5.702	97.574	99.300	1.000	
2	6.007	5.931	102.370	103.960	1.000	
3	5.389	5.626	107.306	99.658	1.000	
4	5.596	6.117	89.667	99.613	1.000	Selected
5	5.893	5.520	111.621	100.645	1.000	
6	6.271	6.250	82.500	100.688	1.000	
7	5.656	6.694	91.741	99.073	1.000	
8	6.341	7.243	112.742	98.502	1.000	
9	5.730	5.686	114.136	102.090	1.000	
10	6.152	7.228	115.833	98.481	1.000	
11	5.844	5.433	106.333	98.959	1.000	
12	6.252	7.151	106.406	99.391	1.000	
13	5.081	5.843	108.694	99.907	1.000	
14	5.030	6.611	114.000	100.483	1.000	

15	6.093	5.653	111.144	102.721	1.000
16	5.592	5.716	113.898	101.859	1.000
17	5.393	5.473	119.882	98.490	1.000
18	6.305	5.832	92.607	102.394	1.000
19	6.177	6.110	89.103	102.353	1.000
20	5.882	6.950	115.393	101.711	1.000
21	5.916	6.783	115.322	103.494	1.000
22	5.661	6.871	113.163	101.658	1.000
23	5.549	6.892	115.536	101.113	1.000
24	5.194	6.135	101.940	100.624	1.000
25	5.858	5.757	111.468	103.088	1.000
26	5.964	6.402	107.100	105.238	1.000
27	5.500	6.087	113.353	103.604	1.000
28	6.112	6.431	90.787	102.468	1.000
29	5.335	5.666	116.941	100.376	1.000
30	5.048	6.019	110.989	100.778	1.000
31	6.318	6.104	87.108	102.330	1.000
32	5.567	6.233	113.570	104.180	1.000
33	5.044	5.635	116.174	98.648	1.000
34	5.754	6.531	97.018	102.046	1.000
35	5.076	6.648	111.590	100.357	1.000
36	5.413	6.126	89.951	98.675	1.000
37	5.929	5.851	89.965	100.258	1.000
38	5.996	5.560	115.418	101.449	1.000
39	6.338	5.799	100.824	103.853	1.000
40	4.731	6.121	106.630	98.432	1.000
41	4.770	6.051	114.776	99.368	1.000
42	5.390	5.760	107.879	100.899	1.000
43	6.229	5.845	82.603	99.043	1.000
44	5.709	5.747	101.550	101.322	1.000
45	5.776	5.437	110.774	99.170	1.000
46	5.001	6.054	107.184	100.151	1.000
47	5.320	6.558	113.010	102.276	1.000
48	5.614	5.521	112.930	99.864	1.000
49	6.058	6.860	92.634	99.793	1.000
50	6.000	5.794	104.223	103.232	1.000
51	6.257	5.638	91.406	100.046	1.000
52	6.167	5.789	98.726	102.896	1.000
53	4.713	6.479	109.251	98.488	1.000
54	5.077	6.868	116.781	99.169	1.000
55	5.564	6.489	101.210	102.204	1.000

56	5.811	6.820	117.839	102.831	1.000
57	6.086	5.745	89.961	100.106	1.000
58	6.006	6.202	81.749	99.011	1.000
59	5.394	5.812	94.765	98.738	1.000
60	6.035	6.765	104.440	103.183	1.000
61	5.871	7.079	107.142	99.408	1.000
62	5.372	6.477	116.010	103.006	1.000
63	5.457	6.735	112.813	101.906	1.000
64	5.950	7.182	119.614	98.774	1.000
65	5.277	5.749	110.415	100.508	1.000
66	6.328	7.064	113.747	101.377	1.000
67	5.532	5.786	96.510	99.733	1.000
68	5.717	5.474	117.248	99.629	1.000
69	5.502	5.920	94.463	99.870	1.000
70	6.062	6.465	103.446	104.973	1.000
71	5.801	7.124	118.284	99.264	1.000
72	5.267	6.145	99.082	100.453	1.000
73	6.222	5.867	83.843	99.629	1.000
74	5.622	5.658	107.289	100.968	1.000
75	6.352	5.628	115.931	102.888	1.000
76	5.082	6.070	119.857	101.484	1.000
77	6.158	6.615	112.010	105.422	1.000
78	5.438	6.303	114.745	103.610	1.000
79	5.872	6.059	98.178	103.205	1.000
80	5.802	6.699	104.658	102.794	1.000
81	5.922	6.149	87.921	100.825	1.000
82	6.109	6.794	103.413	103.044	1.000
83	5.502	6.009	92.135	99.585	1.000
84	5.169	6.336	109.883	101.766	1.000
85	4.999	6.303	109.308	100.694	1.000
86	4.986	6.189	108.387	100.475	1.000
87	5.607	6.352	100.245	102.596	1.000
88	5.612	6.524	119.893	103.958	1.000
89	5.836	5.647	109.438	101.830	1.000
90	5.362	5.707	107.094	100.237	1.000
91	5.709	6.293	92.992	101.375	1.000
92	6.278	6.469	80.322	99.409	1.000
93	5.671	6.657	117.835	103.546	1.000
94	5.611	6.104	97.280	101.863	1.000
95	4.851	5.976	116.206	99.706	1.000
96	6.237	6.321	105.373	106.190	1.000

97	6.096	6.421	115.484	106.119	1.000
98	5.559	5.722	101.996	100.529	1.000
99	5.753	7.008	105.457	99.666	1.000
100	6.082	6.721	93.346	101.410	1.000

Appendix C: Fit Statistics and Lack of Fit Tests

Presents the model validation data, including R^2 , adjusted R^2 and predicted R^2 , and ANOVA significance tests confirming the model adequacy.

Std. Dev.	2.19	R²	0.9966
Mean	64.56	Adjusted R²	0.9907
C.V. %	3.39	Predicted R²	0.8830
		Adeq Precision	45.5821

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Linear	3355.26	11	305.02	107.63	< 0.0001	
2FI	3245.11	8	405.64	143.14	< 0.0001	
Quadratic	249.80	5	49.96	17.63	0.0034	Suggested
Cubic	10.06	1	10.06	3.55	0.1183	Aliased
Pure Error	14.17	5	2.83			

Appendix D: FTIR Absorption Peaks of Enzymatic Hydrolysate

Lists the wavenumbers, absorbances, and corresponding functional group assignments of glucose obtained from enzymatic hydrolysis, confirming structural identity.

Peak No.	Wavenumber (cm ⁻¹)	Absorbance (a.u.)	Assignment / Interpretation
1	3347.0	0.966	O–H stretching (hydroxyl groups of glucose)
2	2919.0	0.199	C–H stretching (aliphatic CH)
3	2850.1	0.116	C–H stretching (aliphatic CH)
4	1450.3	0.101	CH ₂ bending / deformation
5	1370.5	0.090	C–H bending / deformation
6	1150.0	0.545	C–O–C / C–O stretching (glycosidic bond region)
7	1080.1	0.604	C–O stretching (pyranose ring vibrations)
8	1030.0	0.499	C–O stretching / O–H bending
9	950.1	0.280	Anomeric C–O–C stretching (reducing sugar indicator)
10	880.3	0.158	α/β -anomeric C–H deformation (glucose confirmation)

Appendix E: Experimental Results for Kinetics of Enzymatic Hydrolysis

Shows the experimental data for substrate concentration, reaction time, and glucose concentration used for kinetic modeling.

Substrate (mg/mL)	Time (min)	Absorbance (540nm)	Glucose Conc. (mg/mL)	Substrate Remaining (mg/mL)
5	0	0.000	0.0228	5
	5	0.080	0.6588	4.4025
	10	0.160	1.2949	3.8260
	15	0.230	1.8662	3.3214
	20	0.300	2.4374	2.8179
10	0	0.000	0.0228	10
	5	0.120	0.9817	9.1133
	10	0.240	1.9395	8.2498
	15	0.340	2.7400	7.5282
	20	0.410	3.2992	7.0232
15	0	0.000	0.0228	15
	5	0.150	1.2218	13.8989
	10	0.290	2.3401	12.8890
	15	0.370	2.9795	12.3115
	20	0.480	3.8586	11.5205

Appendix F: Kinetic Analysis Tables

Substrate (mg/mL)	Time (min)	Substrate Remaining (mg/mL)	$\ln S_0/S$	$1/S$
5	0	5	0	0.2
	5	4.4025	0.12727	0.22714
	10	3.826	0.26762	0.26137
	15	3.3214	0.40905	0.30108
	20	2.8179	0.57345	0.35487
10	0	10	0	0.1
	5	9.1133	0.09285	0.10973
	10	8.2498	0.1924	0.12122
	15	7.5282	0.28393	0.13283
	20	7.0232	0.35337	0.14239
15	0	15	0	0.06667
	5	13.8989	0.07624	0.07195
	10	12.889	0.15168	0.07759
	15	12.3115	0.19752	0.08122
	20	11.5205	0.26392	0.0868

Initial Substrate Concentration S_0 (mg/mL)	Initial Rate r_A (mg/mL·min)	$\ln S_0$	$\ln r_A$
5	0.1195	1.6094	-2.1244
10	0.1773	2.3026	-1.7299
15	0.2202	2.7081	-1.5137

S_0 (mg/mL)	r_A (mg/mL·min)	$1/S_0$	$1/r_A$
5	0.1195	0.2	8.3682
10	0.1773	0.1	5.64016
15	0.2202	0.06667	4.54133